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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Hydrocarbons

JEE Main Chemistry · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	160 questions
Numerical	28 questions
Answer key	at the end

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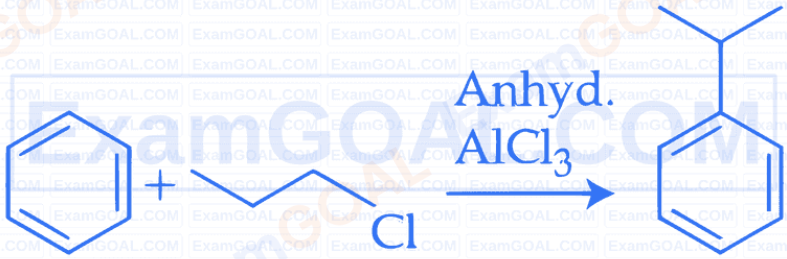
MCQ (Single Correct Answer) 160 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

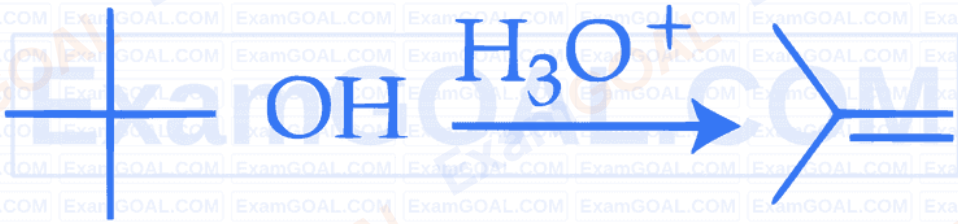
Open solution

The major product of which of the following reaction is not obtained by rearrangement reaction?

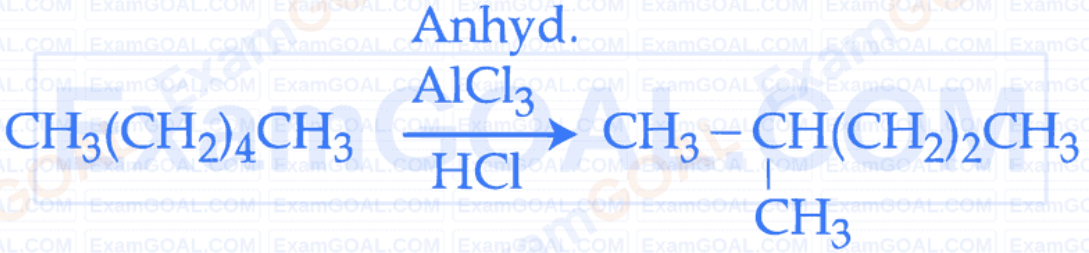
(A)



(B)



(C)



(D)

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Open solution

n-Butane on monochlorination under photochemical condition gives an optically active compound " P ". " P " on further chlorination gives dichloro compounds.

The number of dichloro compounds obtained (ignore stereoisomers) is :

(A) 3 (B) 4 (C) 5 (D) 6

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Q3. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

Given below are two statements :

Statement I : Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of CH_3MgBr with water.

Statement II : Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

IUPAC name of the some alkenes are given below.

Find out the correct stability order.

A. 2-Methylbut-2-ene

B. cis-But-2-ene

C. 2,3-Dimethylbut-2-ene

D. Prop-1-ene

Choose the correct answer from the options given below :

(A) $C > A > B > D$

(B) $C > A > D > B$

(C) $B > D > A > C$

(D) $A > B > C > D$

Q5. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Identify the correct IUPAC name of hydrocarbon (x) containing three primary carbon atoms and with molar mass $72\ g\ mol^{-1}$.

(A) 1, 1 - Dimethylcyclopropane

(B) 2, 2 - Dimethylpropane

(C) 2 - Methylbutane

(D) n-pentane

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Q6. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

For the following Friedel Craft's alkylation reaction, which of the statements are correct?

A. Major product is n-propyl benzene.

B. iso-propyl carbocation intermediate is also generated.

C. Multiple substitution is inevitable.

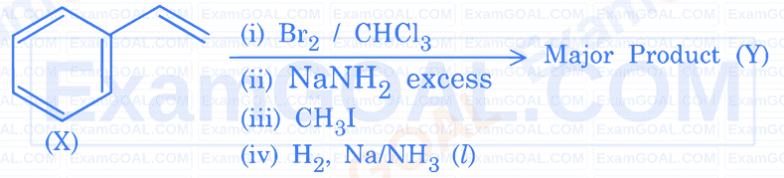
D. Introducing electron-donating substituent on benzene will not produce any alkyl benzene.

Choose the correct answer from the options given below :

(A) A and D only **(B)** B and C only **(C)** A and C only **(D)** B and D only

Q7. JEE Main 2026 (Online) 4th April Evening Shift

Open solution



Compound (X) is subjected to the sequence of reactions as shown above. Molar mass of the major product (Y) formed is ____ gmol^{-1}.

(Given molar mass in $\mathrm{gmol}^{-1} \mathrm{C}: 12, \mathrm{H}: 1, \mathrm{O}: 16$)

(A) 90 **(B)** 118 **(C)** 160 **(D)** 125

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Q8. JEE Main 2026 (Online) 4th April Morning Shift

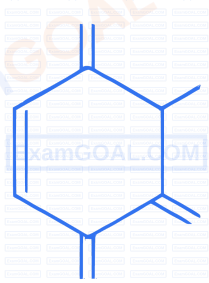
Open solution

An alkene (X) on ozonolysis followed by reduction gives following products.

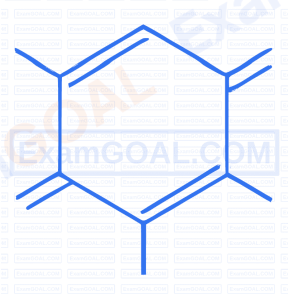
The alkene (X) is :

(A)

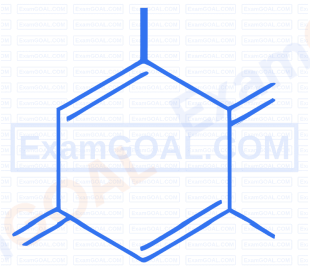
(B)



(C)



(D)



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Q9. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

The compound (X) on

(i) on heating in the presence of anhydrous $AlCl_3$ and HCl gas gives 2,4-dimethyl pentane

(ii) aromatization gives toluene and

(iii) cyclisation gives methyl cyclohexane

The correct name of compound (X) is:

(A) Hept-2-ene

(B) Hept-1,3,5-triene

(C) Heptane

(D) Hept-2,4,6-triene

Q10. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Arrange the following alkenes in decreasing order of stability.

Choose the correct answer from the options given below:

(A) III > I > II > IV

(B) III > II > I > IV

(C) I > III > II > IV

(D) I > III > IV > II

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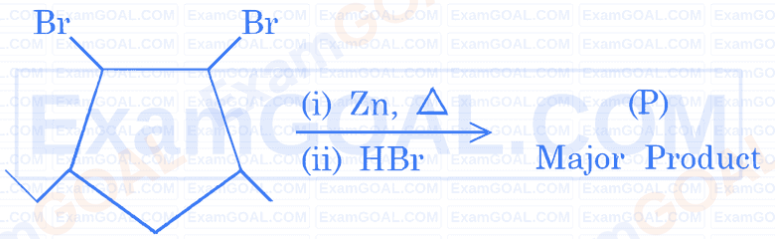
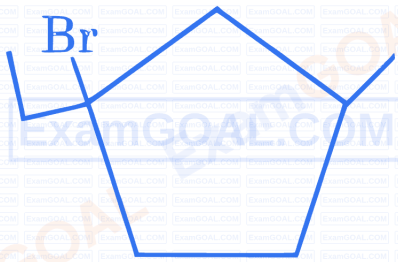
Q11. JEE Main 2026 (Online) 23rd January Evening Shift



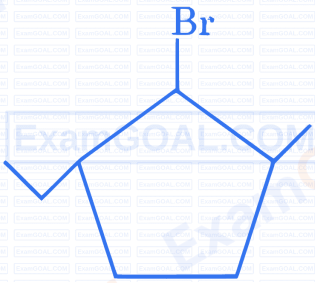
Open solution

Identify (P)

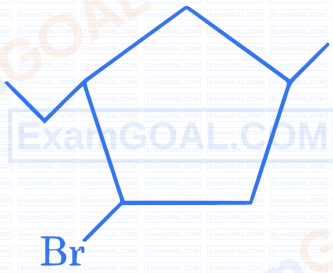
(A)



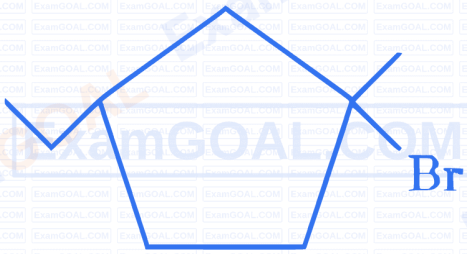
(B)



(C)



(D)



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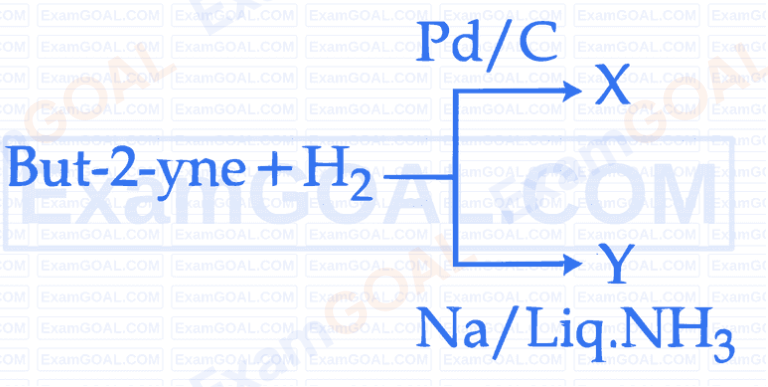
Q12. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

But-2-yne and hydrogen (one mole each) are separately treated with (i) $\mathrm{Pd/C}$ and (ii) $\mathrm{Na/}$ liq. $\mathrm{NH_3}$ to give the products X and Y respectively.



Identify the incorrect statements.

A. X and Y are stereoisomers.

B. Dipole moment of X is zero.

C. Boiling point of X is higher than Y .

D. X and Y react with $\mathrm{O_3/Zn + H_2O}$ to give different products.

Choose the correct answer from the options given below :

(A) B and D Only **(B)** A and C Only **(C)** A and B Only **(D)** B and C Only

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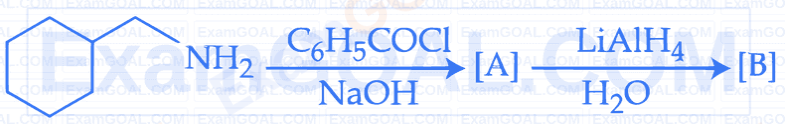
Q13. JEE Main 2026 (Online) 22nd January Evening Shift

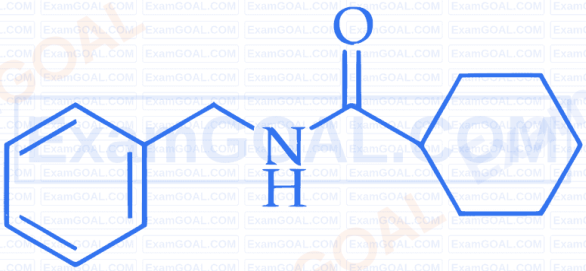


Open solution

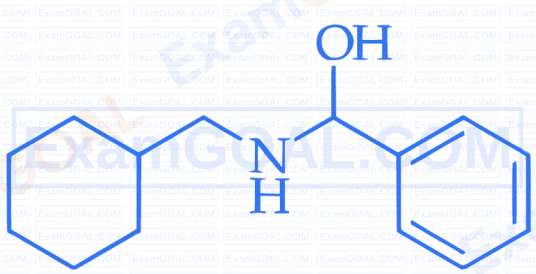
The final product $[B]$ is :

(A)

(B)



(C)



(D)

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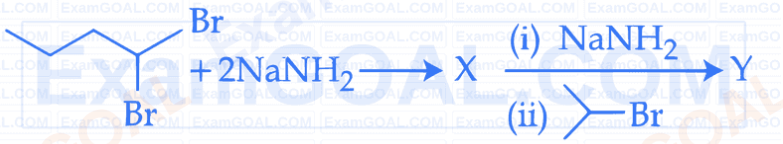
Q14. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Consider the following reaction :



The product Y formed is :

(A) 2-methylhex-3-yne

(B) 5-methylhex-2-yne

(C) Isopropylbut-1-yne

(D) 2-methylhex-2-yne

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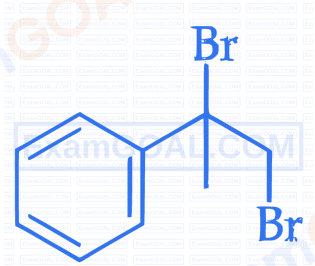
Q15. JEE Main 2026 (Online) 22nd January Evening Shift

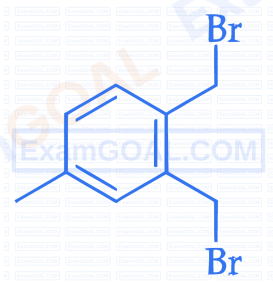
Open solution

The dibromo compound $[P]$ (molecular formula: $C_9H_{10}Br_2$) when heated with excess sodamide followed by treatment with dilute HCl gives [Q]. On warming [Q] with mercuric sulphate and dilute sulphuric acid yield $[R]$ which gives positive Iodoform test but negative Tollen's test. The compound $[P]$ is :

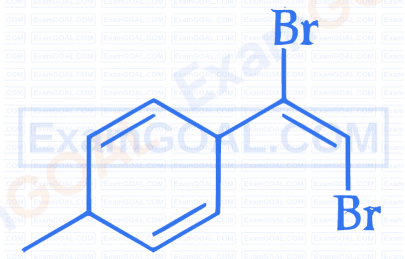
(A)



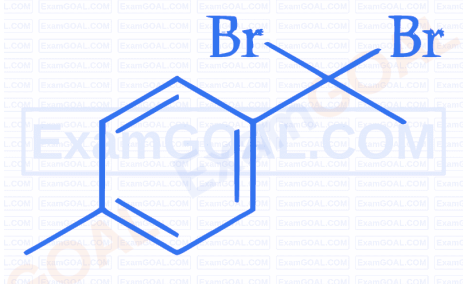
(B)



(C)



(D)



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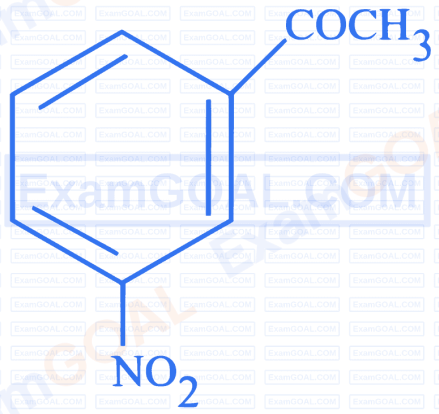
Q16. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Given below are two statements :

Statement I : Benzene is nitrated to give nitrobenzene, which on further treatment with $CH_3COCl/AlCl_3$ will give



Statement II : $-NO_2$ group is a m-directing, and deactivating group.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

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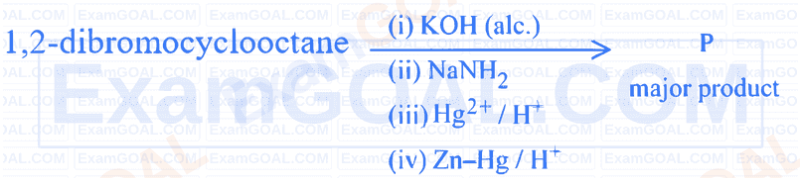
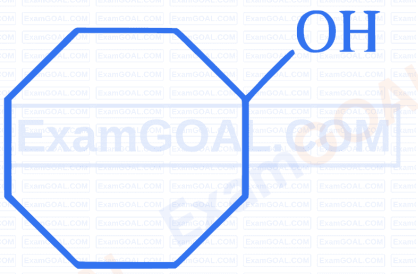
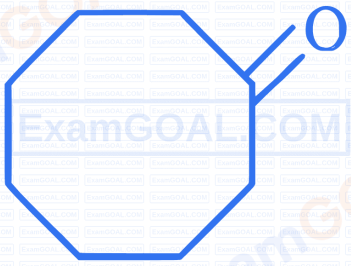
Q17. JEE Main 2025 (Online) 8th April Evening Shift



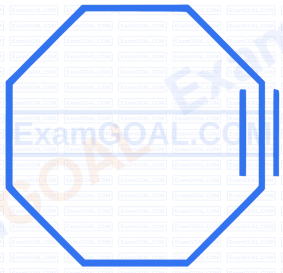
Open solution

'P' is

(A) 

(B) 

(C)

(D) 

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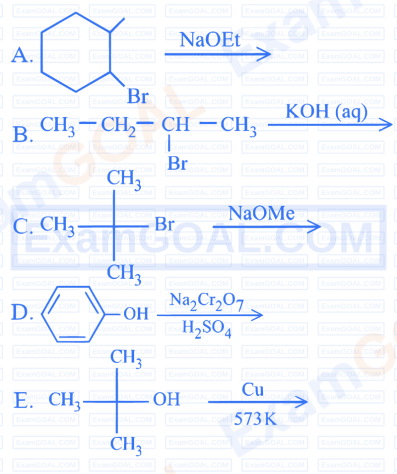
Q18. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

The reactions which cannot be applied to prepare an alkene by elimination, are



Choose the correct answer from the options given below:

(A) B & E Only **(B)** B, C & D Only **(C)** A, C & D Only **(D)** B & D Only

Q19. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Given below are two statements:

Statement I : Ozonolysis followed by treatment with Zn, H_2O of cis-2-butene gives ethanal.

Statement II : The product obtained by ozonolysis followed by treatment with Zn, H_2O of 3, 6-dimethyloct-4-ene has no chiral carbon atom.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are False

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

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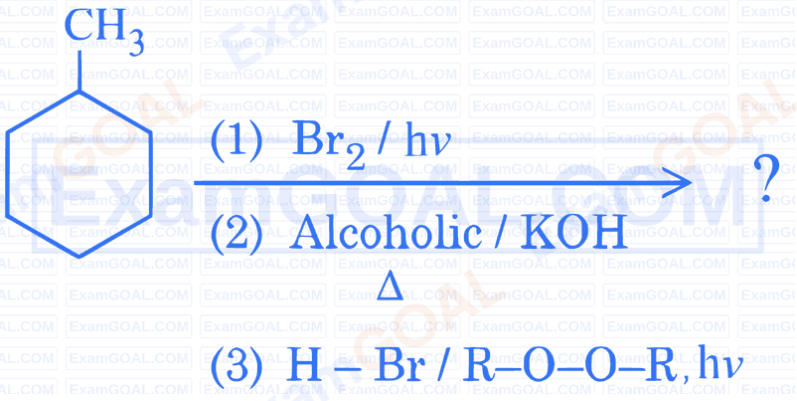
Q20. JEE Main 2025 (Online) 4th April Morning Shift

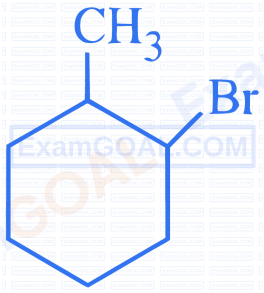
Open solution

Predict the major product of the following reaction sequence:-



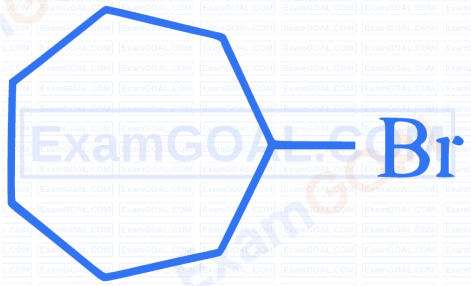
(A)

(B)



(C)

(D)



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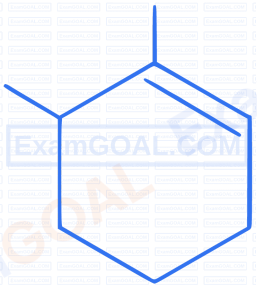
Q21. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

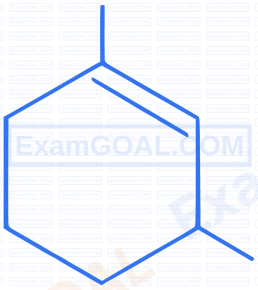
Which compound would give 3-methyl-6-oxoheptanal upon ozonolysis?

(A)

(B)



(C)



(D)

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Q22. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Given below are two statements :

Statement (I) : Neopentane forms only one monosubstituted derivative.

Statement (II) : Melting point of neopentane is higher than n-pentane.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Statement I is correct but Statement II is incorrect

(D) Both Statement I and Statement II are incorrect

Q23. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Match List - I with List - II.

List - I (Reaction)		List - II (Name of reaction)	
(A)		(I)	Lucas reaction
(B)	$ArN_2^+X^- \xrightarrow[HCl]{Cu} ArCl + N_2 \uparrow + CuX$	(II)	Finkelstein reaction
(C)	$C_2H_5Br + NaI \xrightarrow[Acetone]{Dry} C_2H_5I + NaBr$	(III)	Fittig reaction
(D)	$CH_3C(OH)(CH_3)CH_3 \xrightarrow[ZnCl_2]{HCl} CH_3C(Cl)(CH_3)CH_3$	(IV)	Gatterman reaction

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(B) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(C) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q24. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Total number of sigma (σ) _____ and pi(π) _____ bonds respectively present in hex-1-en-4-yne are:

(A) 14 and 3 **(B)** 11 and 3 **(C)** 13 and 3 **(D)** 3 and 13

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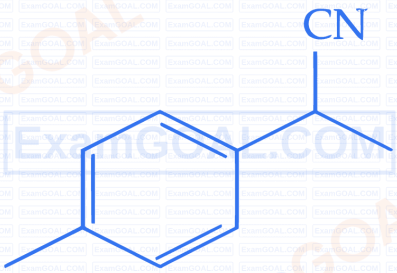
Q25. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

The product B formed in the following reaction sequence is:

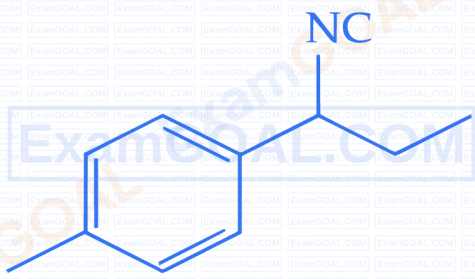
(A)

(B)



(C)

(D)



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Q26. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Identify product [A], [B] and [C] in the following reaction sequence.

$$CH_3 - C \equiv CH \xrightarrow[H_2]{Pd/C} [A] \xrightarrow[(ii)\ Zn,H_2O]{(i)\ O_3} [B] + [C]$$

(A) [A]: CH_3–CH=CH_2, [B]: CH_3CHO, [C]: CH_3CH_2OH

(B) [A]: CH_3–CH=CH_2, [B]: CH_3CHO, [C]: HCHO

(C) [A]: $CH_3CH_2CH_3$, [B]: CH_3CHO, [C]: HCHO

(D)

$$[A] : CH_2 = CH_2,\ [B] : H_3C - \overset{\overset{O}{\|}}{C} - CH_3,\ [C] : HCHO$$

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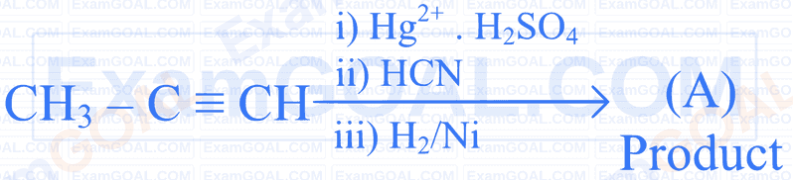
Q27. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

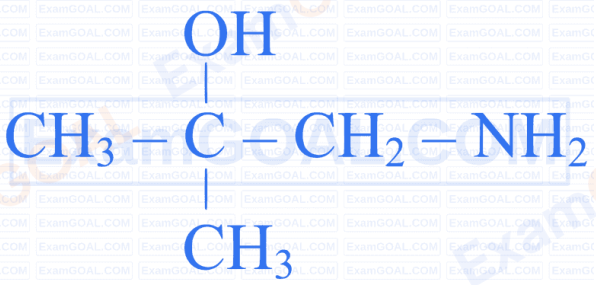
The product (A) formed in the following reaction sequence is



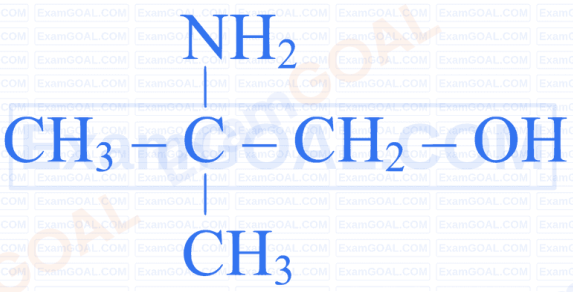
(A)

(B)

(C)



(D)



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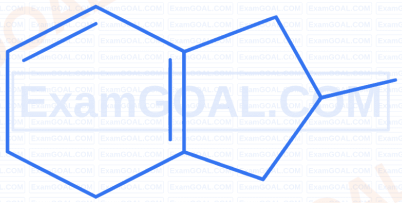
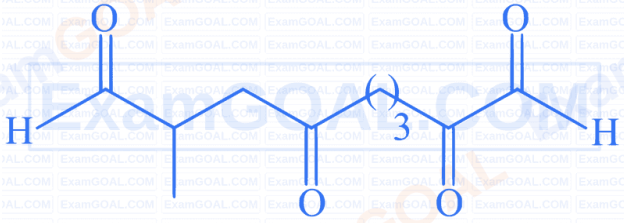
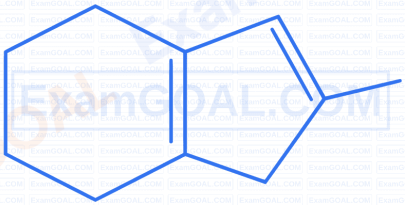
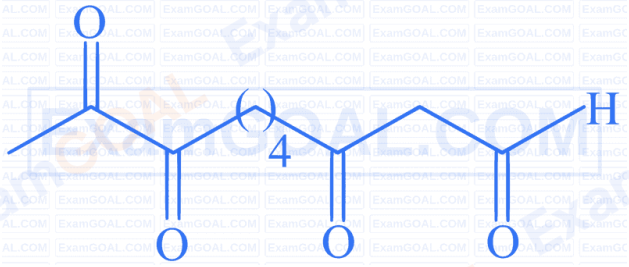
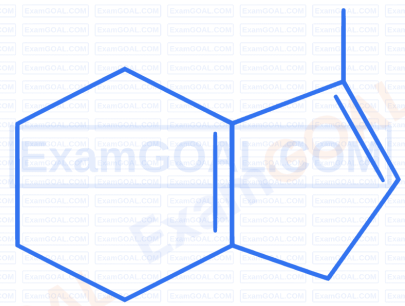
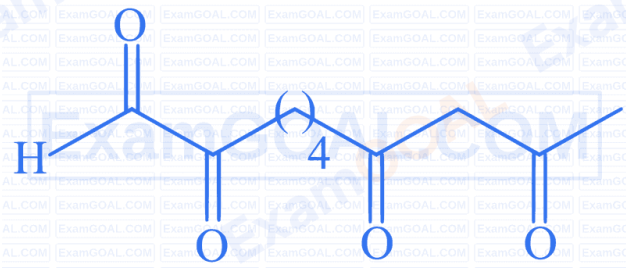
Q28. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Match List - I with List - II

	List - I (Isomers of $C_{10}H_{14}$)		List - II (Ozonolysis product)
(A)		(I)	
(B)		(II)	
(C)		(III)	
(D)		(IV)	

Choose the correct answer from the options given below :

(A) (A) $-$(III), (B) $-$ (IV), (C) $-$ (I), (D)$-$ (II)

(B) (A) $-$ (I), (B) $-$ (IV), (C) $-$ (III), (D) $-$ (II)

(C) (A) $-$ (III), (B) $-$ (II), (C) $-$ (I), (D) $-$ (IV)

(D) (A) $-$ (II), (B) $-$ (II), (C) $-$ (I), (D) $-$ (IV)

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Q29. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Match the List - I with List - II

	List - I **Name reaction**		**List - II** **Product obtainable**
(A)	Swarts reaction	(I)	Ethyl benzene
(B)	Sandmeyer's reaction	(II)	Ethyl iodide
(C)	Wurtz Fittig reaction	(III)	Cyanobenzene
(D)	Finkelstein reaction	(IV)	Ethyl fluoride

Choose the correct answer from the options given below:

(A) A-II, B-III, C-I, D-IV

(B) A-II, B-I, C-III, D-IV

(C) A-IV, B-I, C-III, D-II

(D) A-IV, B-III, C-I, D-II

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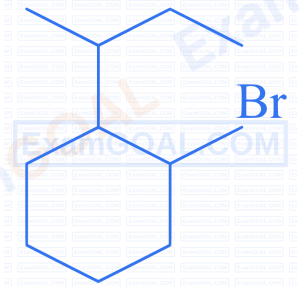
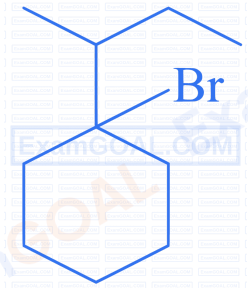
Q30. JEE Main 2025 (Online) 22nd January Evening Shift

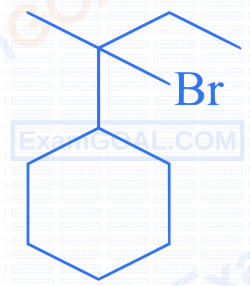
Open solution

When sec-butylcyclohexane reacts with bromine in the presence of sunlight, the major product is :

(A) 

(B) 

(C)

(D) 

Q31. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

The alkane from below having two secondary hydrogens is :

(A) 2,2,4,5-Tetramethylheptane

(B) 2,2,4,4-Tetramethylhexane

(C) 4-Ethyl-3,4-dimethyloctane

(D) 2,2,3,3-Tetramethylpentane

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Q32. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Given below are two statements :

Statement I : One mole of propyne reacts with excess of sodium to liberate half a mole of H_2 gas.

Statement II : Four g of propyne reacts with $NaNH_2$ to liberate NH_3 gas which occupies 224 mL at STP.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are correct

(D) Statement I is correct but Statement II is incorrect

Q33. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

The incorrect statement regarding ethyne is

(A) The carbon-carbon bond in ethyne is weaker than that in ethene

(B) Both carbons are sp hybridised

(C) The $C-C$ bond in ethyne is shorter than that in ethene

(D) Ethyne is linear

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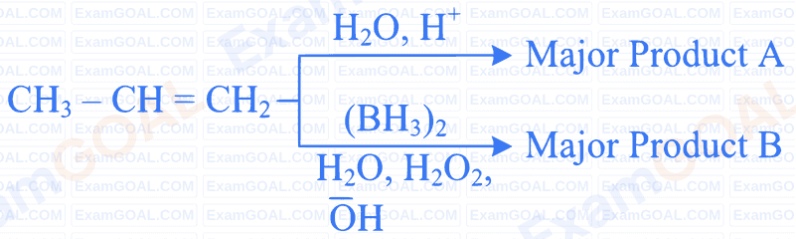
Q34. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Identify the product A and product B in the following set of reactions.



(A) A - $CH_3CH_2CH_2-OH$ B - $CH_3CH(OH)-CH_3$

(B) A - $CH_3-CH(OH)-CH_3$ B - $CH_3CH_2CH_2-OH$

(C) A - $CH_3CH_2CH_2-OH$ B - $CH_3CH_2CH_2-OH$

(D) A - $CH_3CH_2CH_3$ B - $CH_3CH_2CH_3$

Q35. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

In the given compound, the number of 2° carbon atom/s is _____.

(A) Two (B) Three (C) Four (D) One

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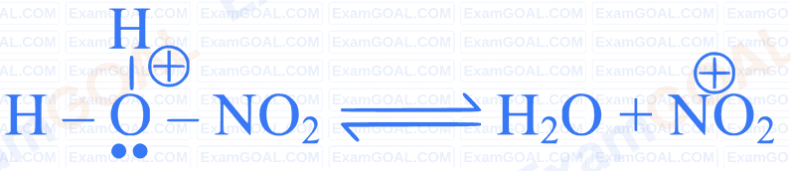
Q36. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

Given below are two statements :

Statement I : Nitration of benzene involves the following step -



Statement II : Use of Lewis base promotes the electrophilic substitution of benzene.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are incorrect

(B) Statement I is correct but Statement II is incorrect

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are correct

Q37. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

In the above chemical reaction sequence "A" and "B" respectively are

(A) H_2O, H^+ and $NaOH_{(alc)}/I_2$

(B) $O_3, Zn/H_2O$ and $KMnO_4$

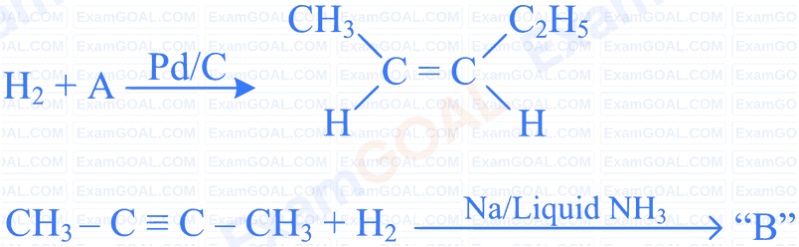
(C) H_2O, H^+ and $KMnO_4$

(D) $O_3, Zn/H_2O$ and $NaOH_{(alc)}/I_2$

Q38. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

In the given reactions identify A and B



(A) A : n-Pentane B : Cis - 2 - butene

(B) A : 2-Pentyne B : Cis-2-butene

(C) A : n-Pentane B : trans-2-butene

(D) A : 2-Pentyne B : trans-2-butene

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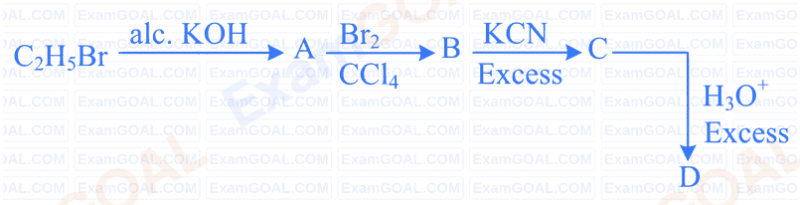
Q39. JEE Main 2024 (Online) 1st February Evening Shift



Open solution

Acid **D** formed in above reaction is :

(A) Malonic acid (B) Oxalic acid (C) Succinic acid (D) Gluconic acid

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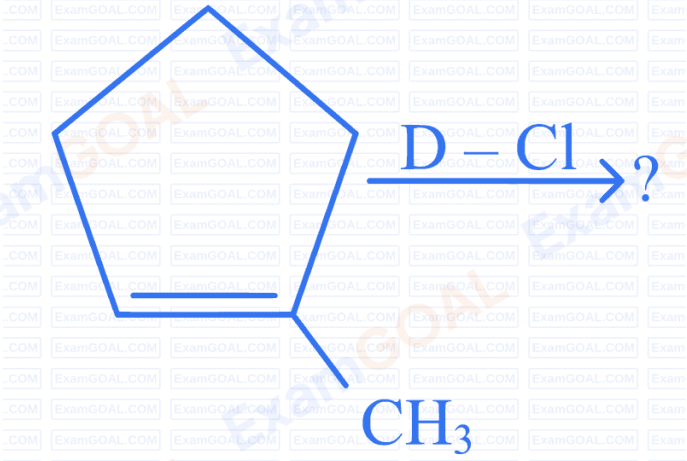
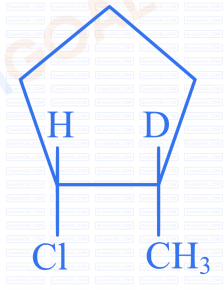
Q40. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

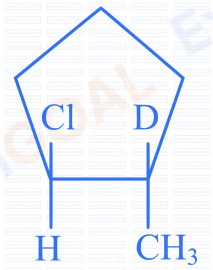
Major product of the following reaction is -



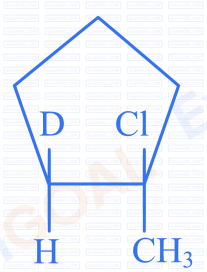
(A)



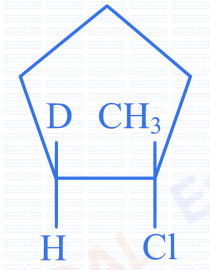
(B)



(C)



(D)



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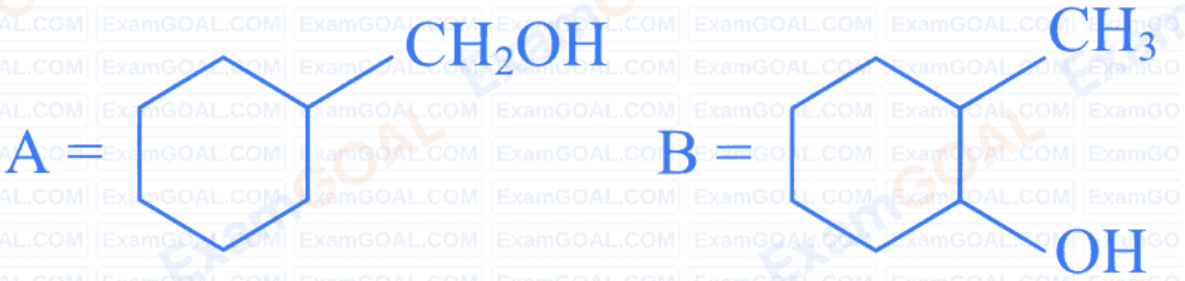
Q41. JEE Main 2024 (Online) 30th January Evening Shift

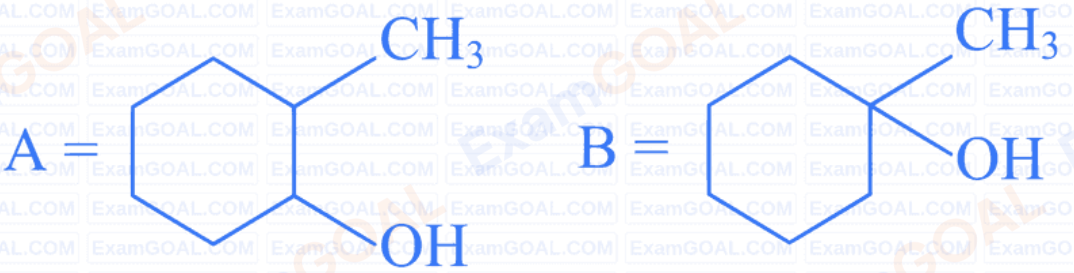
Open solution

Products A and B formed in the following set of reactions are

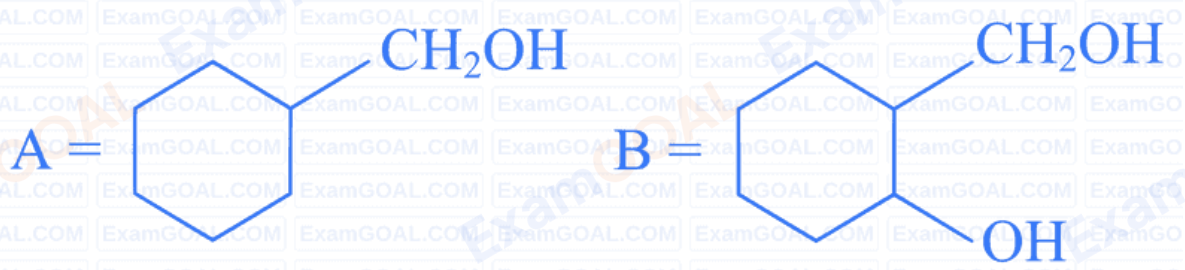
(A)



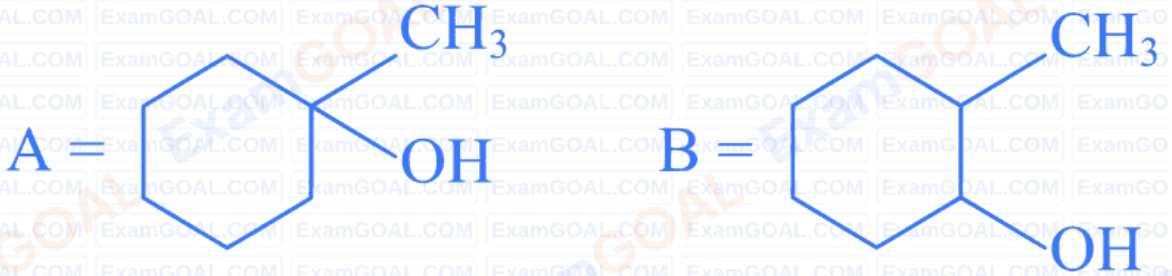
(B)



(C)



(D)



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Q42. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Compound A formed in the following reaction reacts with B to give the product C. Find out A and B.

(A) $A = CH_3 - C \equiv \bar{C}\ Na^+, B = CH_3 - CH_2 - CH_2 - Br$

(B) $A = CH_3 - CH = CH_2,\ B = CH_3 - CH_2 - CH_2 - Br$

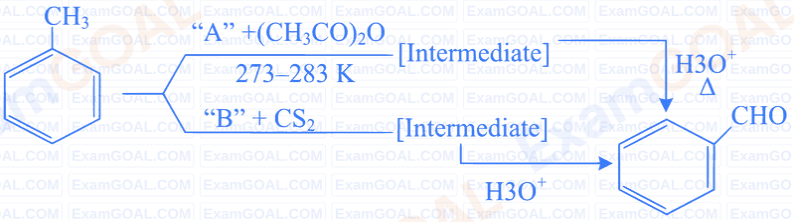
(C) $A = CH_3 - C \equiv \bar{C}\ Na^+, B = CH_3 - CH_2 - CH_3$

(D) $A = CH_3 - CH_2 - CH_3,\ B = CH_3 - C \equiv CH$

Q43. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

In the given reactions, identify the reagent A and reagent B.



(A) $A - CrO_3 \quad B - CrO_3$

(B) $A - CrO_3 \quad B - CrO_2Cl_2$

(C) A- $CrO_2Cl_2 \quad B - CrO_3$

(D) $A - CrO_2Cl_2 \quad B - CrO_2Cl_2$

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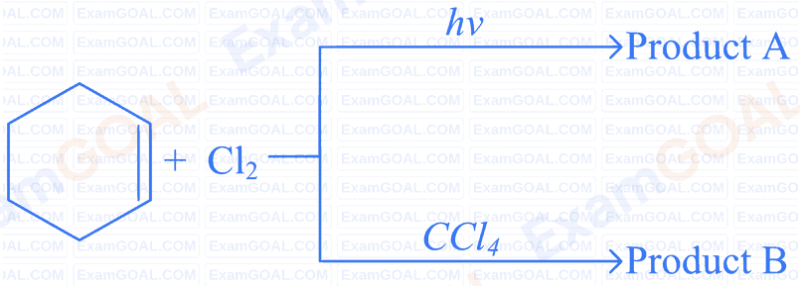
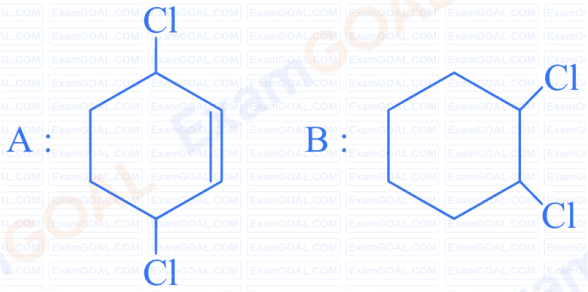
Q44. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

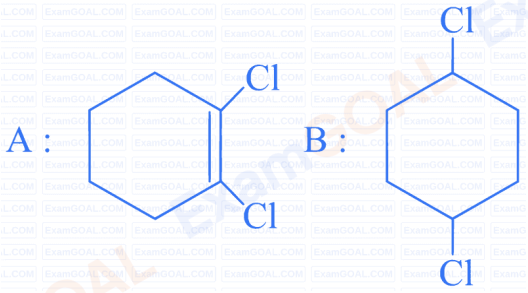
Identify product A and product B :



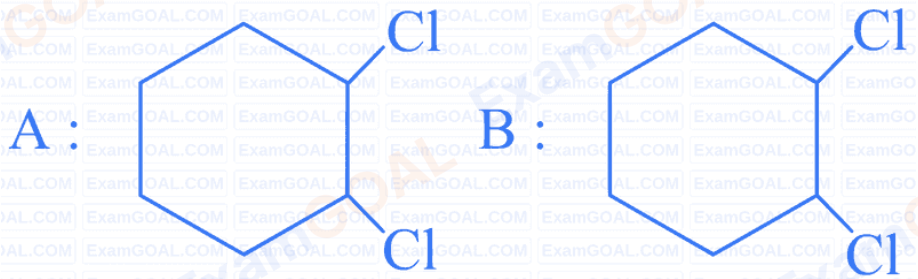
(A)



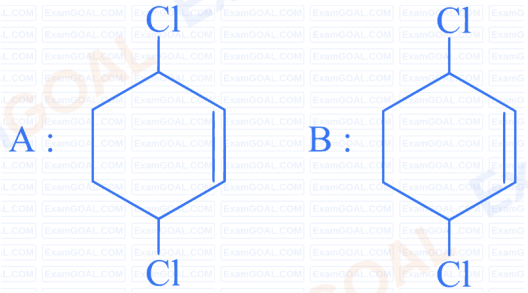
(B)



(C)



(D)



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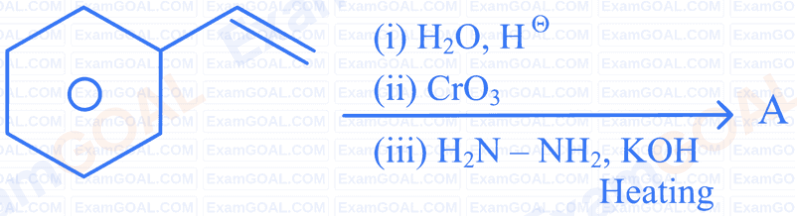
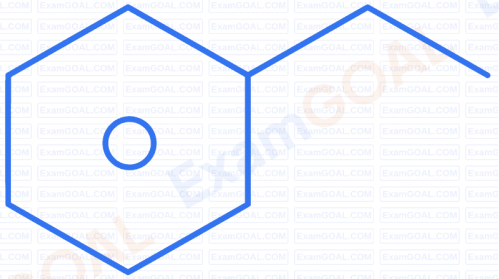
Q45. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

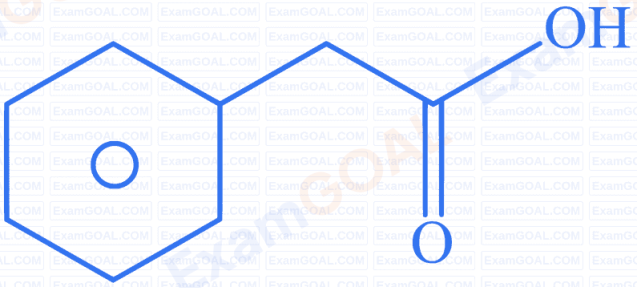
The final product A formed in the following multistep reaction sequence is



(A)

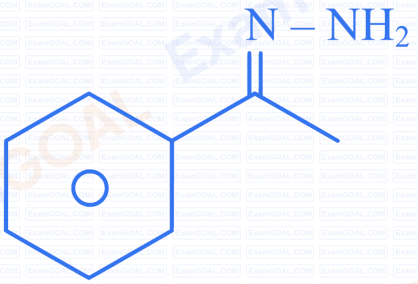


(B)



(C)

(D)



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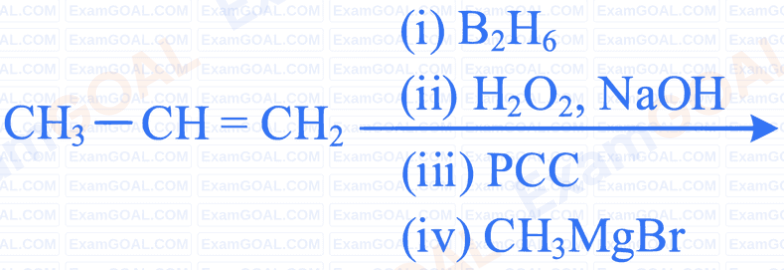
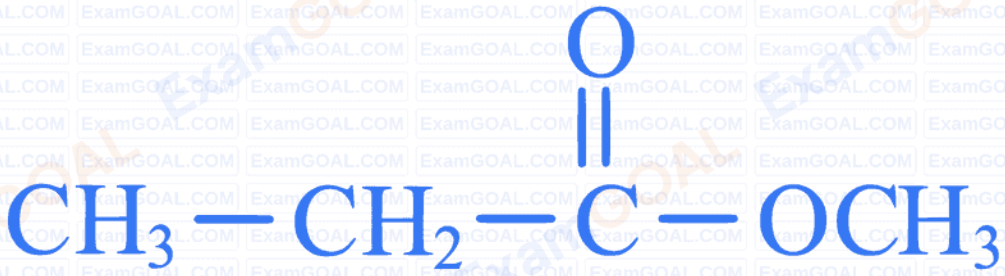
Q46. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

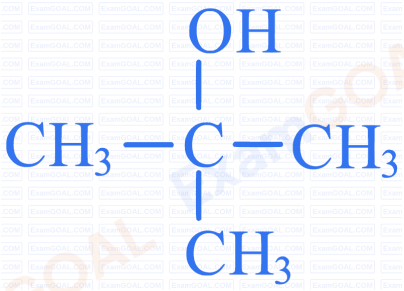
The product formed in the following multistep reaction is :



(A)

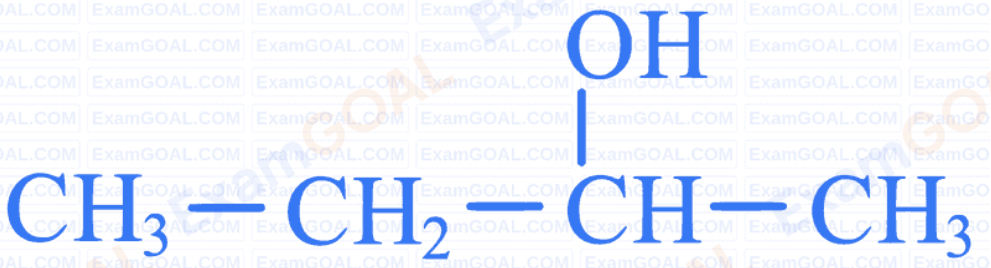


(B)



(C) $CH_3 - CH_2 - CH_2 - CH_2 - OH$

(D)



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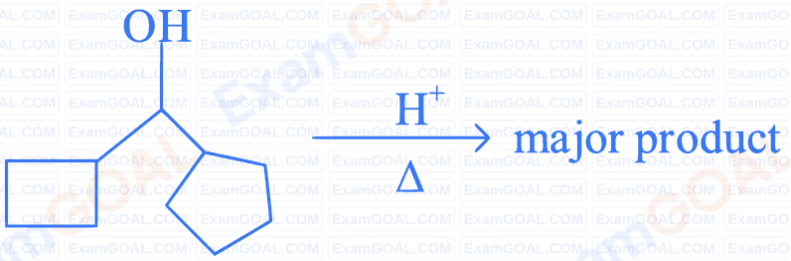
Q47. JEE Main 2023 (Online) 13th April Morning Shift



Open solution

In the above reaction, left hand side and right hand side rings are named as 'A' and 'B' respectively. They undergo ring expansion. The correct statement for this process is:

(A) Finally both rings will become six membered each.

(B) Only A will become 6 membered.

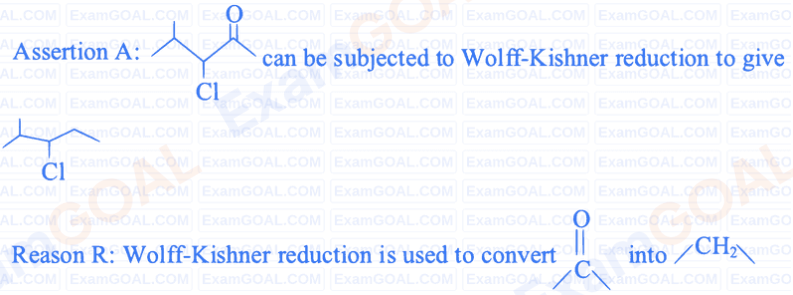
(C) Ring expansion can go upto seven membered rings

(D) Finally both rings will become five membered each.

Q48. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.



In the light of the above statements, choose the correct answer from the options given below:

(A) Both A and R are true and R is the correct explanation of A

(B) A is false but R is true

(C) A is true but R is false

(D) Both A and R are true but R is NOT the correct explanation of A

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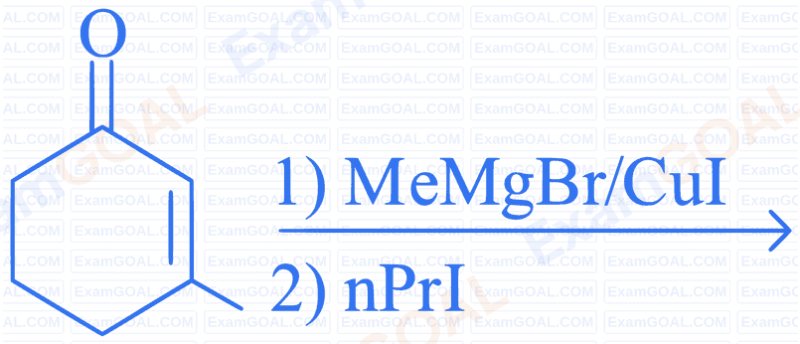
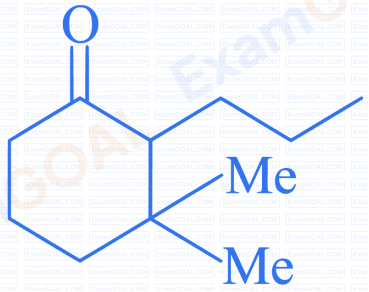
Q49. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

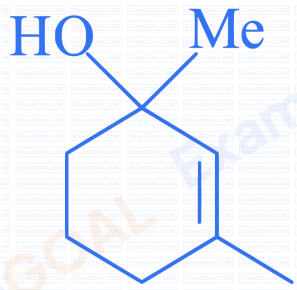
Find out the major product from the following reaction.



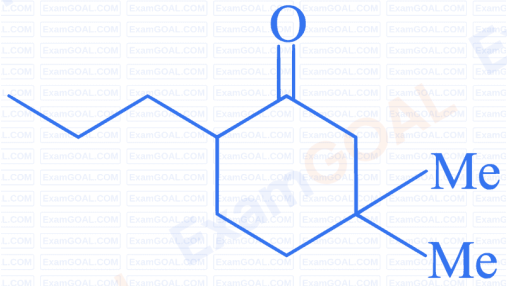
(A)



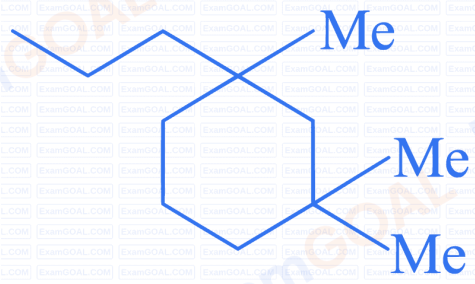
(B)



(C)



(D)



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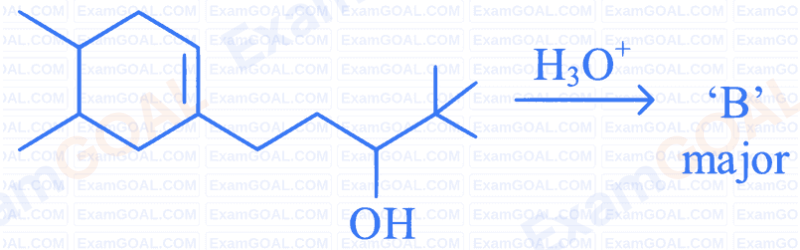
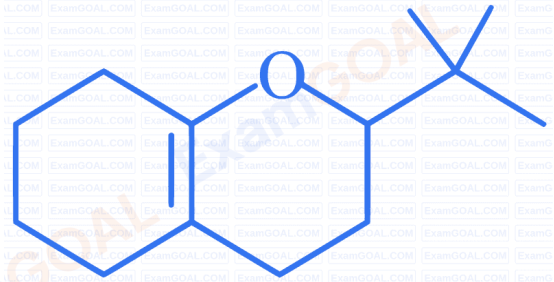
Q50. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

In the following reaction, 'B' is



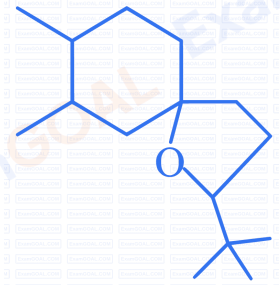
(A)



(B)

(C)

(D)



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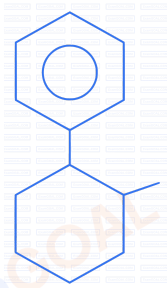
Q51. JEE Main 2023 (Online) 1st February Evening Shift

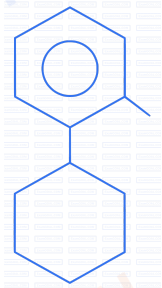
Open solution

'X' is :

(A)



(B)



(C)

(D)

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Q52. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

But-2-yne is reacted separately with one mole of Hydrogen as shown below :

$$\underline{B} \xleftarrow[\text{liq } NH_3]{Na} CH_3-C\equiv C-CH_3 \xrightarrow[\Delta]{Pd/C} \underline{A}$$
$$+H_2$$

A. A is more soluble than B.

B. The boiling point & melting point of A are higher and lower than B respectively.

C. A is more polar than B because dipole moment of A is zero.

D. Br_2 adds easily to B than A.

Identify the **incorrect** statements from the options given below :

(A) A and B only **(B)** A, C & D only **(C)** B, C & D only **(D)** C and D only

Q53. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

A hydrocarbon ' X ' with formula C_6H_8 uses two moles of H_2 on catalytic hydrogenation of its one mole. On ozonolysis, ' X ' yields two moles of methane dicarbaldehyde. The hydrocarbon ' X ' is:

(A) hexa-1, 3, 5-triene **(B)** cyclohexa-1, 4-diene

(C) cyclohexa - 1,3 - diene **(D)** 1-methylcyclopenta-1, 4-diene

Q54. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

Choose the correct set of reagents for the following conversion.

trans $(Ph-CH=CH-CH_3) \rightarrow$ cis $(Ph-CH=CH-CH_3)$

(A) $Br_2, alc \cdot KOH, NaNH_2, H_2$ Lindlar Catalyst **(B)** $Br_2, aq \cdot KOH, NaNH_2, Na$ (Liq NH_3)

(C) $Br_2, alc \cdot KOH, NaNH_2, Na$ (Liq NH_3) **(D)** $Br_2, aq \cdot KOH, NaNH_2, H_2$ Lindlar Catalyst

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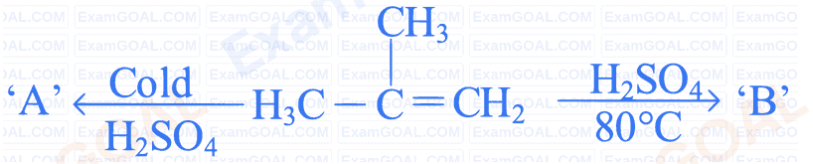
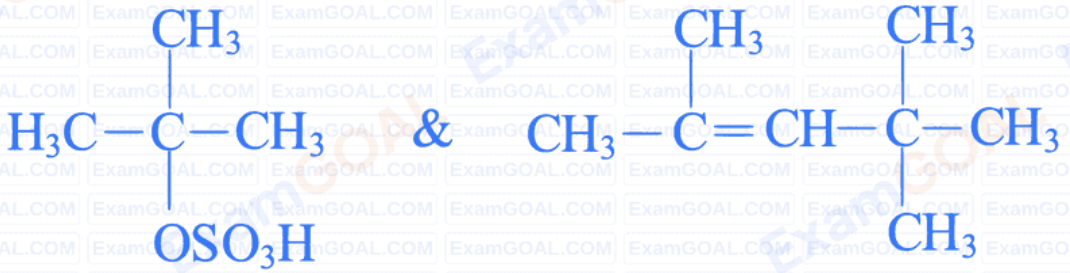
Q55. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

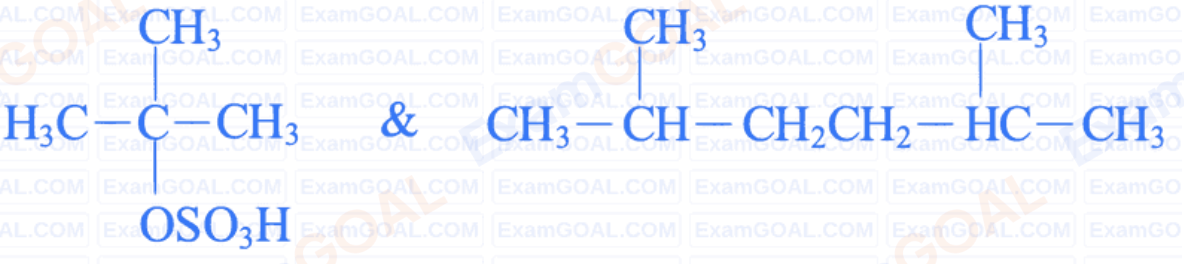
The major products 'A' and 'B', respectively, are



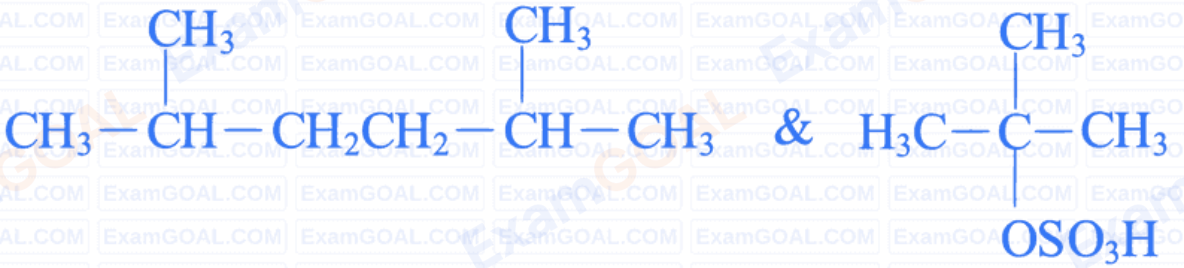
(A)



(B)



(C)



(D)

Q56. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

The one giving maximum number of isomeric alkenes on dehydrohalogenation reaction is (excluding rearrangement)

(A) 2-Bromo-3,3-dimethylpentane

(B) 2-Bromopropane

(C) 1-Bromo-2-methylbutane

(D) 2-Bromopentane

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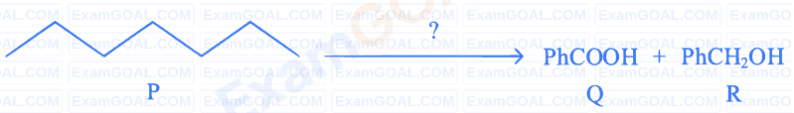
Q57. JEE Main 2023 (Online) 25th January Morning Shift



Open solution

The correct sequence of reagents for the preparation of Q and R is :

(A) $(i)Cr_2O_3, 770K, 20\ atm; (ii)CrO_2Cl_2, H_3O^+; (iii)NaOH; (iv)H_3O^+$

(B) $(i)CrO_2Cl_2, H_3O^+; (ii)Cr_2O_3, 770K, 20\ atm; (iii)NaOH; (iv)H_3O^+$

(C) $(i)KMnO_4, OH^-; (ii)Mo_2O_3, \Delta; (iii)NaOH; (iv)H_3O^+$

(D) $(i)Mo_2O_3, \Delta; (ii)CrO_2Cl_2, H_3O^+; (iii)NaOH; (iv)H_3O^+$

Q58. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

Find out the major products from the following reactions.

(A)

(B)

(C)

(D)

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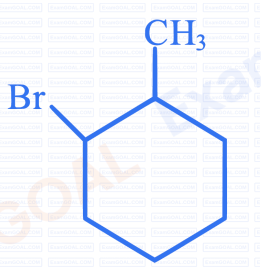
Q59. JEE Main 2023 (Online) 24th January Morning Shift

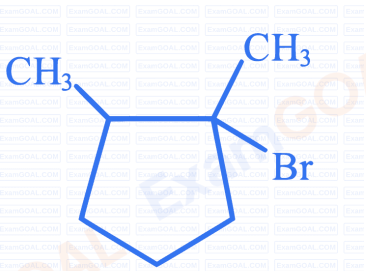
Open solution

In the following given reaction, 'A' is

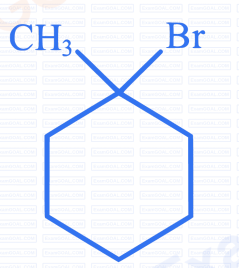
(A)



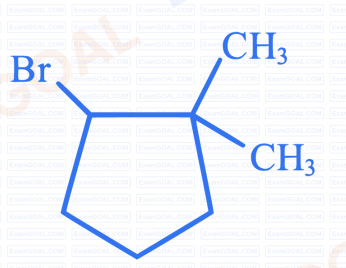
(B)



(C)



(D)



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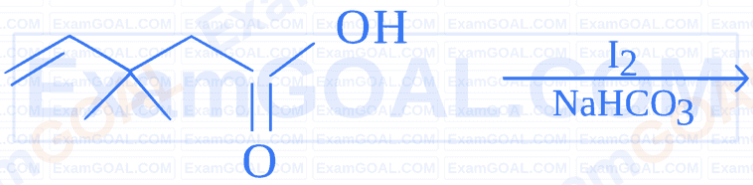
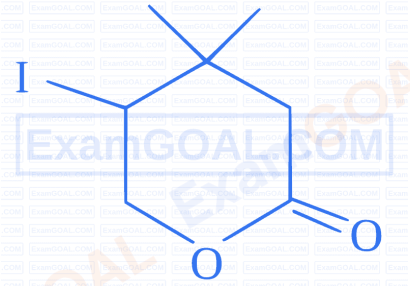
Q60. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

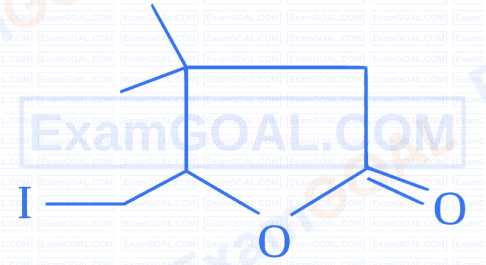
Major product of the following reaction is



(A)



(B)



(C)

(D)

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Q61. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

$\dot{\mathrm{Cl}} + \mathrm{CH_4} \rightarrow \mathrm{A} + \mathrm{B}$

A and B in the above atmospheric reaction step are :

(A) $\mathrm{C_2H_6}$ and $\mathrm{Cl_2}$

(B) $\dot{\mathrm{C}}\mathrm{HCl_2}$ and $\mathrm{H_2}$

(C) $\dot{\mathrm{C}}\mathrm{H_3}$ and HCl

(D) $\mathrm{C_2H_6}$ and HCl

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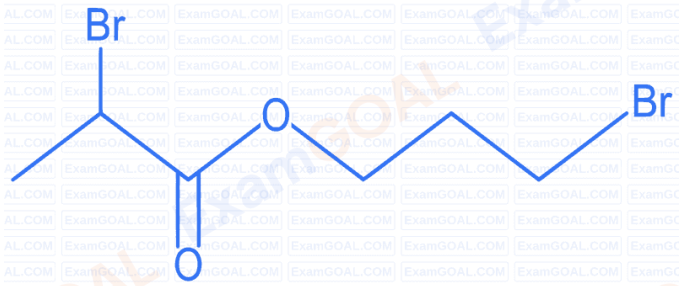
Q62. JEE Main 2022 (Online) 25th July Evening Shift

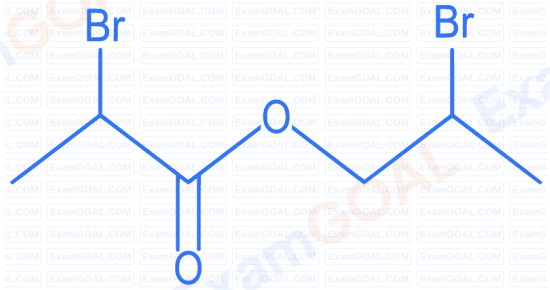
Open solution

Major product of the following reaction is

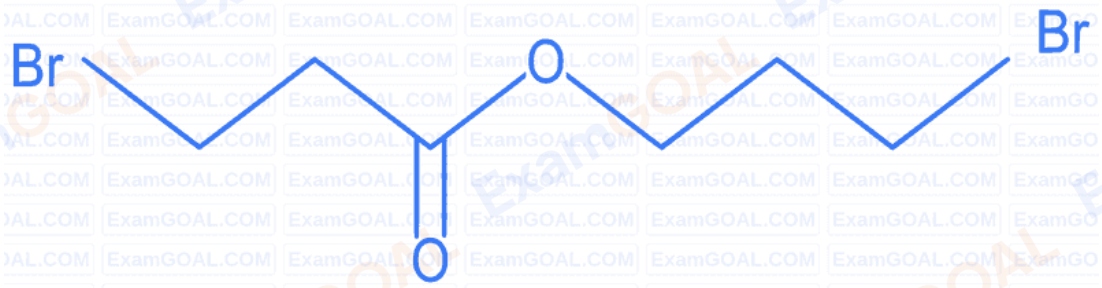
(A)



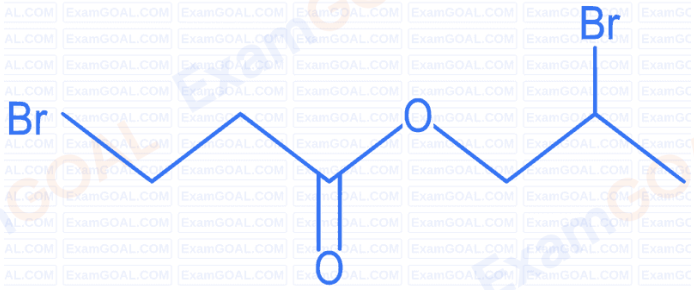
(B)



(C)



(D)



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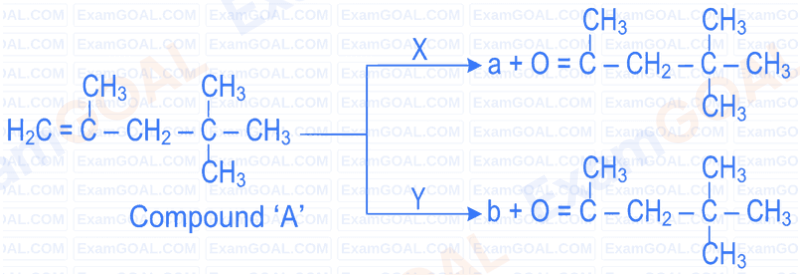
Q63. JEE Main 2022 (Online) 25th July Morning Shift

Open solution

A compound 'A' on reaction with 'X' and 'Y' produces the same major product but different by-products 'a' and 'b'. Oxidation of 'a' gives a substance produced by ants.



'X' and 'Y' respectively are

(A) $KMnO_4/H^+$ and dil. $KMnO_4$, 273 K

(B) $KMnO_4$ (dilute), 273 K and $KMnO_4/H^+$

(C) $KMnO_4/H^+$ and $O_3, H_2O/Zn$

(D) $O_3, H_2O/Zn$ and $KMnO_4/H^+$

Q64. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

Two isomers 'A' and 'B' with molecular formula C_4H_8 give different products on oxidation with $KMnO_4$ in acidic medium. Isomer 'A' on reaction with $KMnO_4/H^+$ results in effervescence of a gas and gives ketone. The compound 'A' is

(A) But-1-ene.

(B) cis-But-2-ene.

(C) trans-But-2-ene.

(D) 2-methyl propene.

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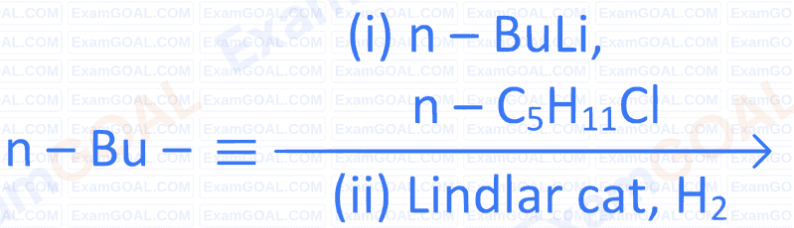
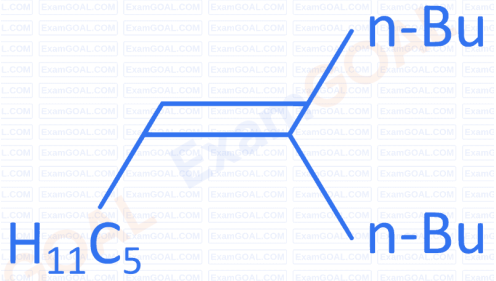
Q65. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

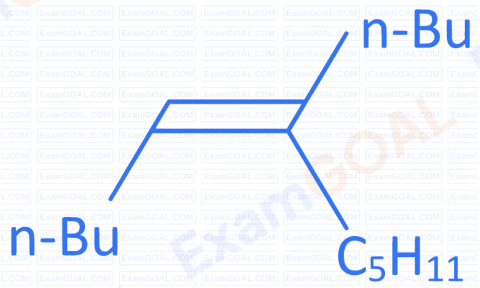
What will be the major product of following sequence of reactions?



(A)

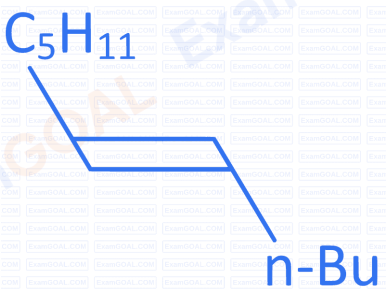


(B)



(C)

(D)



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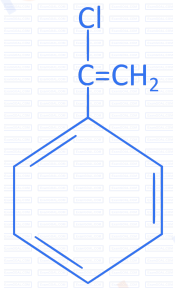
Q66. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

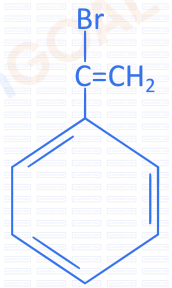
Product 'A' of following sequence of reactions is

$$\text{Ethylbenzene} \xrightarrow[\text{(b) } Cl_2, \Delta]{\text{(a) } Br_2, Fe} \text{'A'} \xrightarrow{\text{(c) alc. KOH}} \text{Major Product}$$

(A)



(B)



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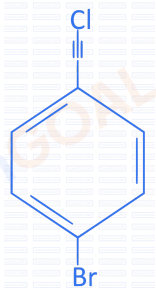

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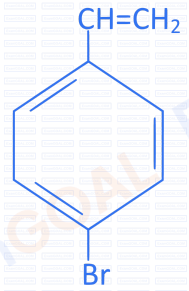

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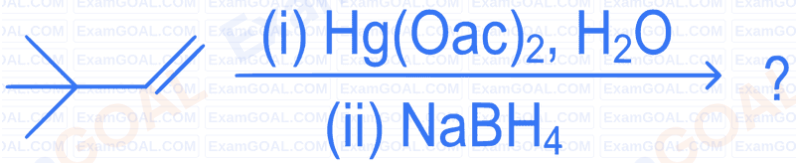
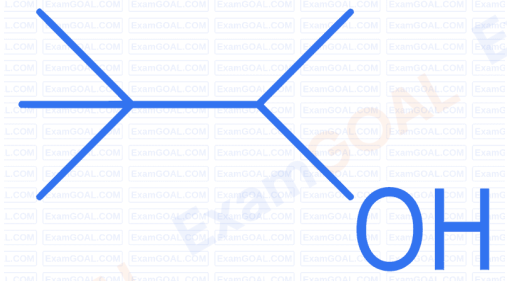
Q67. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

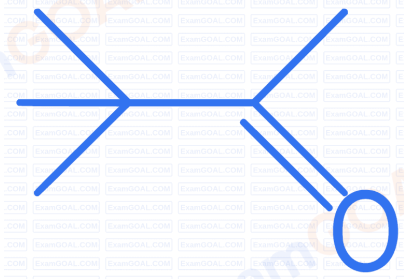
The major product in the following reaction



(A)

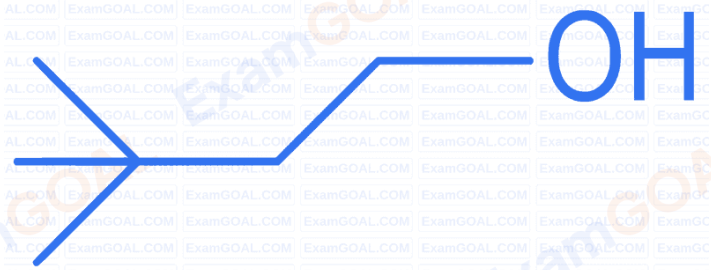


(B)



(C)

(D)



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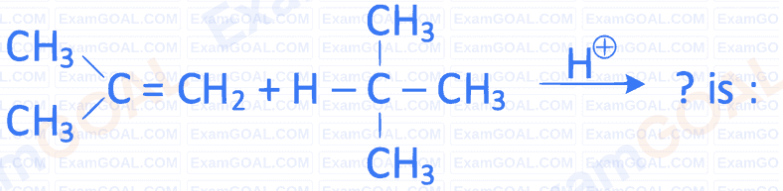
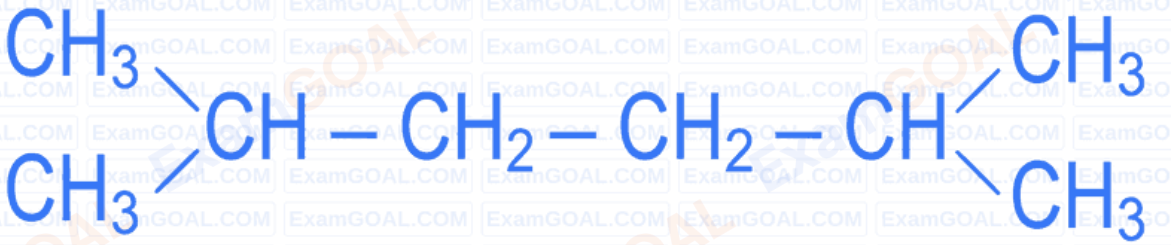
Q68. JEE Main 2022 (Online) 25th June Morning Shift

Open solution

The product formed in the following reaction.



(A)



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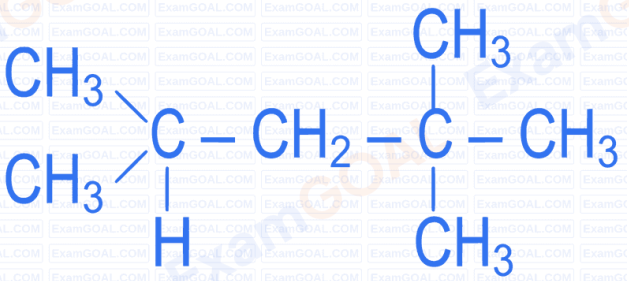
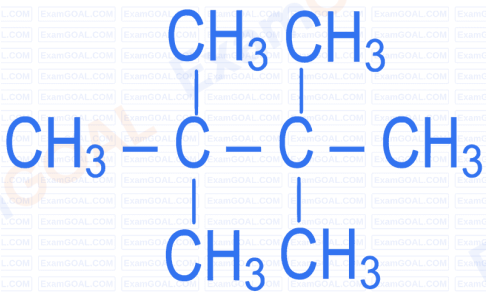


(C)

(D)



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Q69. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

Given below are two statements :

Statement I : The presence of weaker π-bonds makes alkenes less stable than alkanes.

Statement II : The strength of the double bond is greater than that of carbon-carbon single bond.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q70. JEE Main 2022 (Online) 24th June Evening Shift

Open solution

Which of the following reagents / reactions will convert 'A' to 'B' ?

(A) PCC oxidation

(B) Ozonolysis

(C) BH_3, $H_2O_2/^{-}OH$ followed by PCC oxidation

(D) HBr, hydrolysis followed by oxidation by $K_2Cr_2O_7$.

Q71. JEE Main 2021 (Online) 1st September Evening Shift

Open solution

The stereoisomers that are formed by electrophilic addition of bromine to trans-but-2-ene is/are :

(A) 2 enantiomers and 2 mesomers

(B) 2 identical mesomers

(C) 2 enantiomers

(D) 1 racemic and 2 enantiomers

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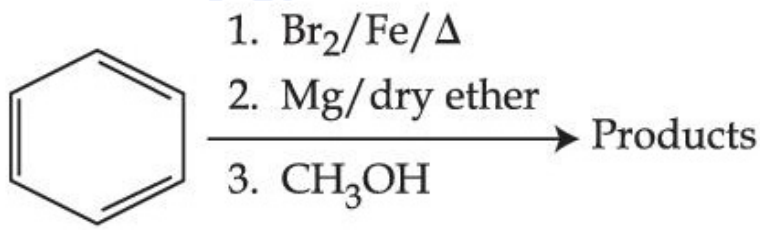
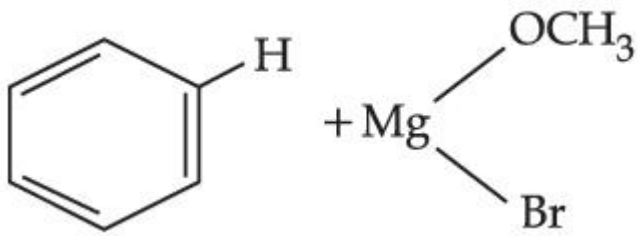
Q72. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

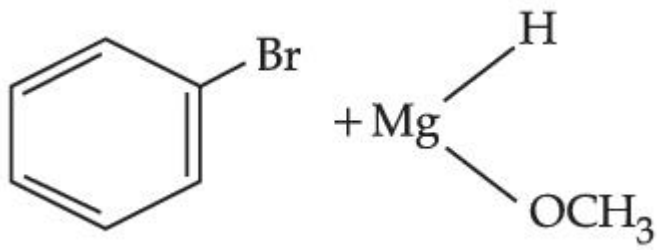
For the following sequence of reactions, the correct products are :



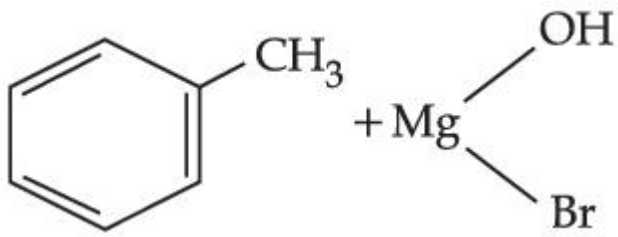
(A)



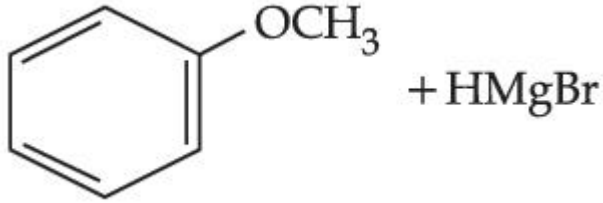
(B)



(C)



(D)



Q73. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

Presence of which reagent will affect the reversibility of the following reaction, and change it to an irreversible reaction :

$$CH_4 + I_2 \underset{\text{Reversible}}{\overset{h\nu}{\rightleftharpoons}} CH_3 - I + HI$$

(A) HOCl

(B) dilute HNO_2

(C) Liquid NH_3

(D) Concentrated HIO_3

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Q74. JEE Main 2021 (Online) 25th July Morning Shift

An organic compound 'A' C_4H_8 on treatment with $KMnO_4/H^+$ yields compound 'B' C_3H_6O.

Compound 'A' also yields compound 'B' on ozonolysis. Compound 'A' is :

Open solution

(A) 2-Methylpropene
(B) 1-Methylcyclopropane
(C) But-2-ene
(D) Cyclobutane

Q75. JEE Main 2021 (Online) 20th July Evening Shift

Metallic sodium does not react normally with :

Open solution

(A) gaseous ammonia
(B) But-2-yne
(C) Ethyne
(D) tert-butyl alcohol

Q76. JEE Main 2021 (Online) 20th July Evening Shift

Benzene on nitration gives nitrobenzene in presence of HNO_3 and H_2SO_4 mixture, where :

Open solution

(A) both H_2SO_4 and HNO_3 act as a bases
(B) HNO_3 acts as an acid and H_2SO_4 acts as a base
(C) both H_2SO_4 and HNO_3 act as an acids
(D) HNO_3 acts as a base and H_2SO_4 acts as an acid

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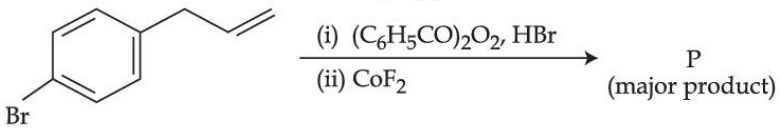
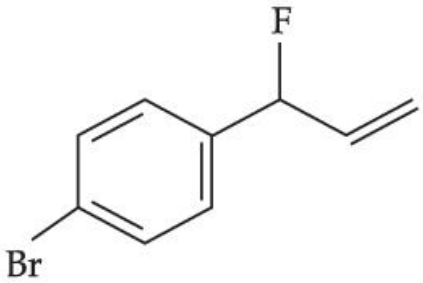
Q77. JEE Main 2021 (Online) 20th July Evening Shift



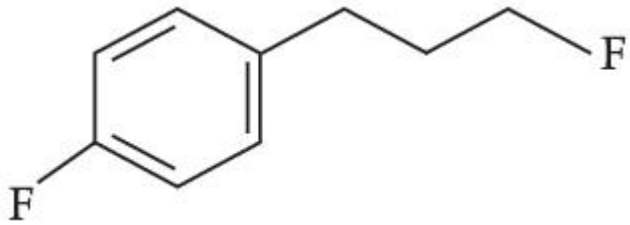
Open solution

Major product P of above reaction, is :

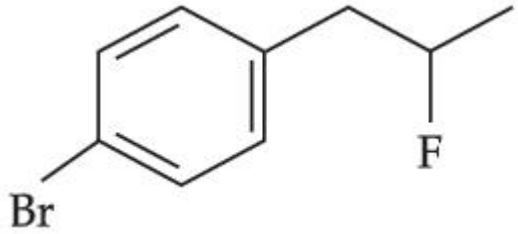
(A)



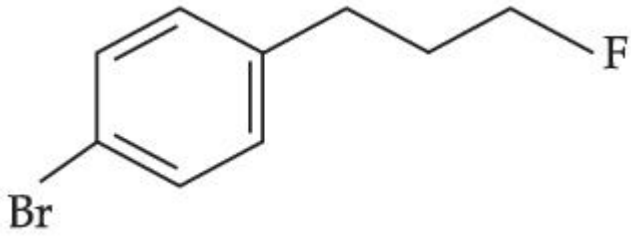
(B)



(C)



(D)



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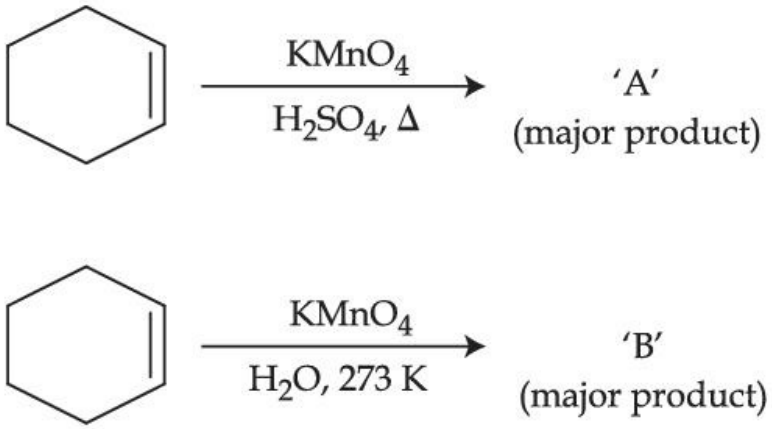
Q78. JEE Main 2021 (Online) 20th July Morning Shift

Open solution



For above chemical reactions, identify the correct statement from the following :

(A) Both compound 'A' and compound 'B' are dicarboxylic acids

(B) Both compound 'A' and compound 'B' are diols

(C) Compound 'A' is diol and compound 'B' is dicarboxylic acid

(D) Compound 'A' is dicarboxylic acid and compound 'B' is diol

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Q79. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

Match List - I with List - II :

List - I (Chemicals)

(a) Alcoholic potassium hydroxide

(b) $Pd/BaSO_4$

(c) BHC (Benzene hexachloride)

(d) Polyacetylene

List - II (Use / Preparation / Constituent)

(i) Electrodes in batteries

(ii) Obtained by addition reaction

(iii) Used for β - elimination reaction

(iv) Lindlar's catalyst

Choose the most appropriate match :

(A) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

(B) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

(C) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

(D) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

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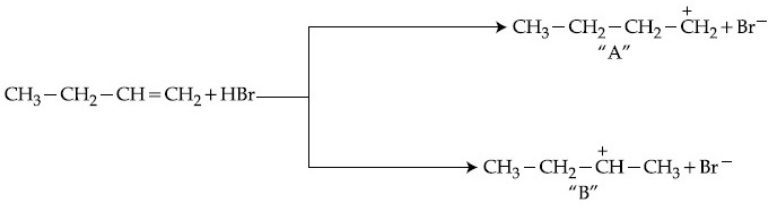
Q80. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

Choose the correct statement regarding the formation of carbocations A and B given.



(A) Carbocation A is more stable and formed relatively at faster rate

(B) Carbocation B is more stable and formed relatively at slow rate

(C) Carbocation B is more stable and formed relatively at faster rate

(D) Carbocation A is more stable and formed relatively at slow rate

Q81. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

Given below are two statements :

Statement I : 2-methylbutane on oxidation with $KMnO_4$ gives 2-methylbutan-2-ol.

Statement II : n-alkanes can be easily oxidised to corresponding alcohols with $KMnO_4$.

Choose the correct option :

(A) Both statement I and statement II are incorrect

(B) Both statement I and statement II are correct

(C) Statement I is correct but statement II is incorrect

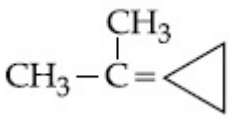
(D) Statement I is incorrect but statement II is correct

Q82. JEE Main 2021 (Online) 16th March Evening Shift

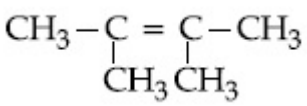
Open solution

An unsaturated hydrocarbon X on ozonolysis gives A. Compound A when warmed with ammonical silver nitrate forms a bright silver mirror along the sides of the test tube. The unsaturated hydrocarbon X is :

(A) $HC \equiv C - CH_2 - CH_3$

(B) 

(C) $CH_3 - C \equiv C - CH_3$

(D) 

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Q83. JEE Main 2021 (Online) 16th March Morning Shift

Which of the following is Lindlar catalyst?

(A) Partially deactivated palladised charcoal

(B) Zinc chloride and HCl

(C) Sodium and Liquid NH_3

(D) Cold dilute solution of $KMnO_4$

Open solution

Q84. JEE Main 2021 (Online) 25th February Evening Shift

The correct sequence of reagents used in the preparation of 4-bromo-2-nitroethyl benzene from benzene is :

(A) $CH_3COCl/AlCl_3, Br_2/AlBr_3, HNO_3/H_2SO_4, Zn/HCl$

(B) $CH_3COCl/AlCl_3, Zn-Hg/HCl, Br_2/AlBr_3, HNO_3/H_2SO_4$

(C) $Br_2/AlBr_3, CH_3COCl/AlCl_3, HNO_3/H_2SO_4, Zn/HCl$

(D) $HNO_3/H_2SO_4, Br_2/AlCl_3, CH_3COCl/AlCl_3, Zn-Hg/HCl$

Open solution

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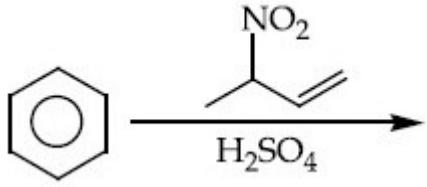
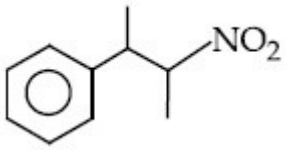
Q85. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

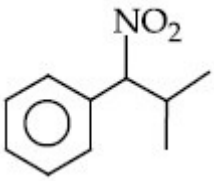
The major product of the following reaction is :



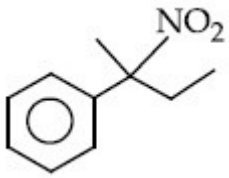
(A)



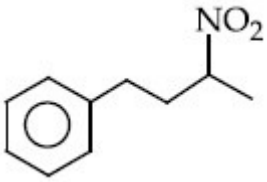
(B)



(C)



(D)



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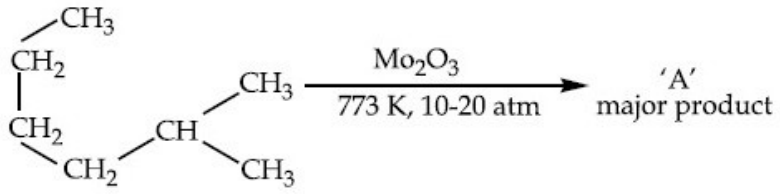
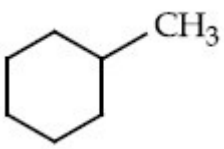
Q86. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

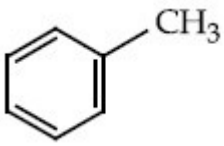
Identify A in the given chemical reaction.



(A) 

(B)

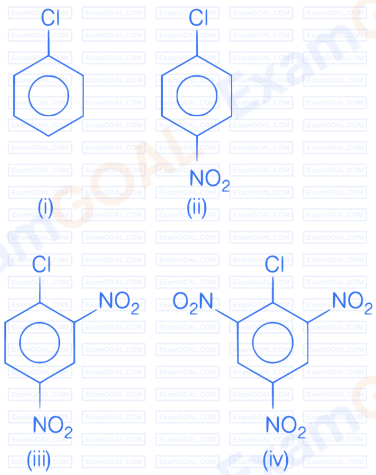
(C)

(D) 

Q87. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

The correct order of the following compounds showing increasing tendency towards nucleophilic substitution reaction is :



(A) (iv) < (i) < (ii) < (iii)

(B) (iv) < (iii) < (ii) < (i)

(C) (iv) < (i) < (iii) < (ii)

(D) (i) < (ii) < (iii) < (iv)

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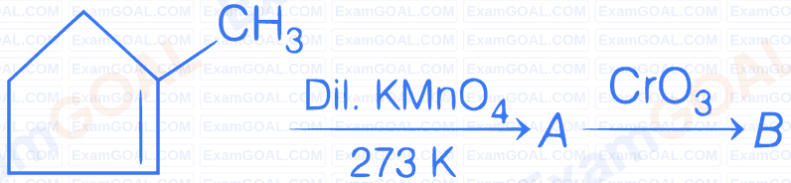
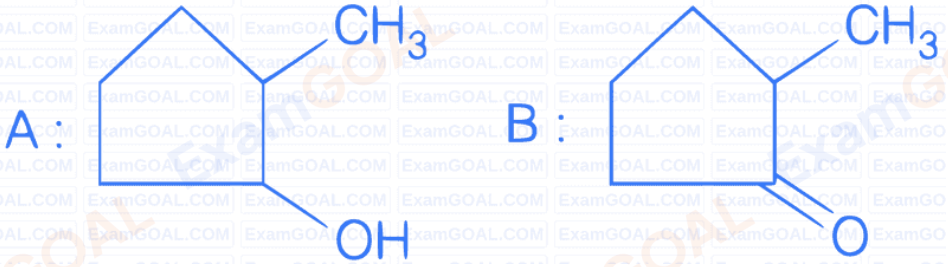
Q88. JEE Main 2021 (Online) 24th February Morning Shift

Identify products A and B.

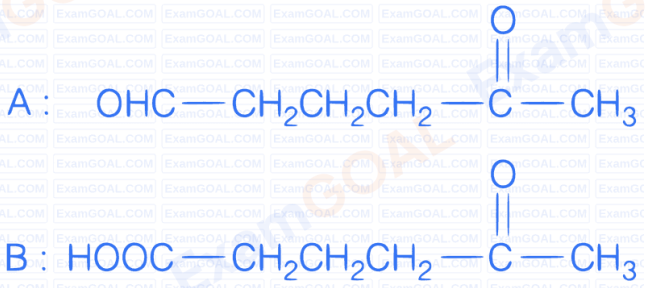
Open solution



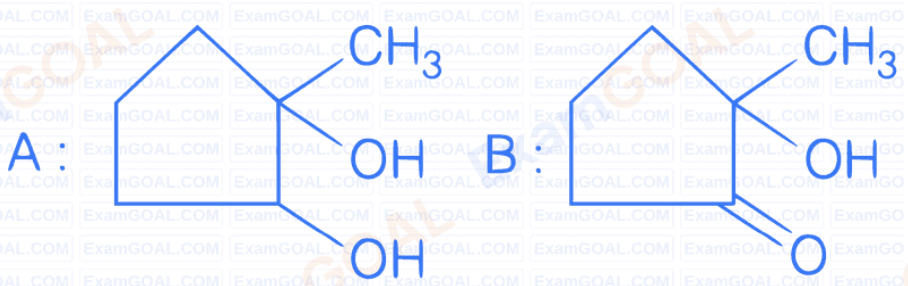
(A)



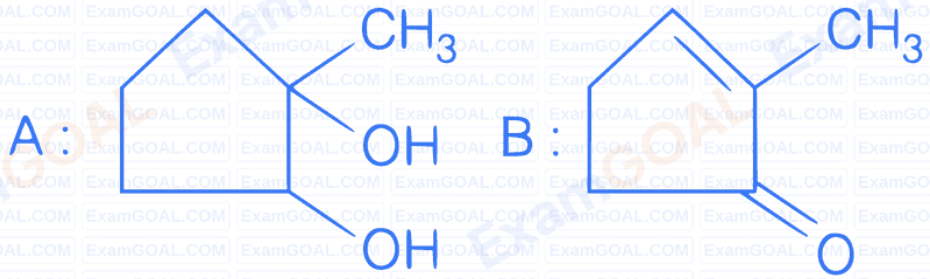
(B)



(C)



(D)



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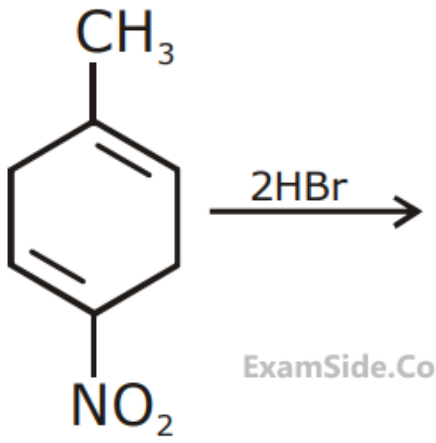
Q89. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

The major product of the following reaction is :



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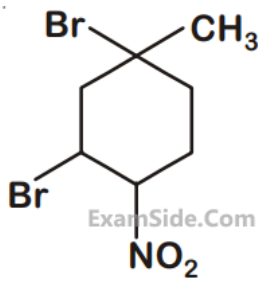
(A)



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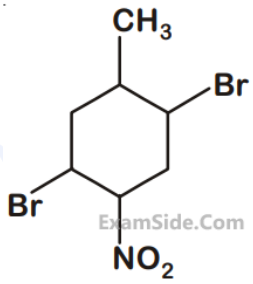
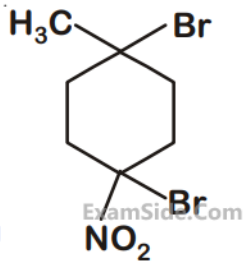
(B)



ExamSide.Com

(C)



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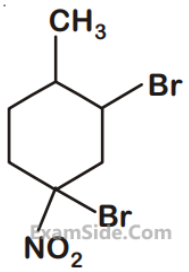
(D)



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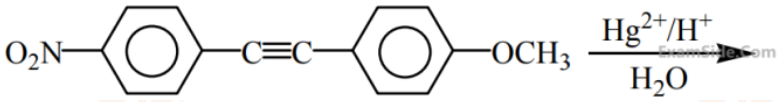
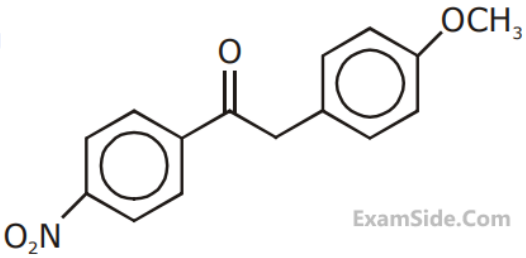
Q90. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

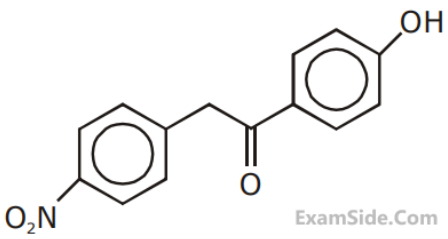
The major product obtained from the following reaction is :



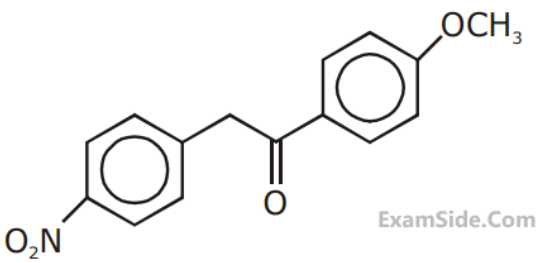
(A)



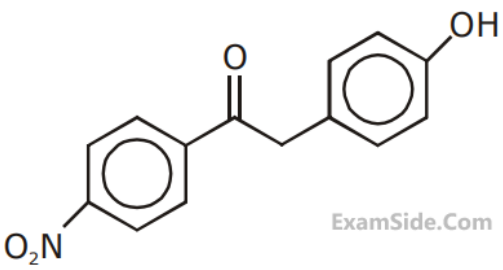
(B)



(C)



(D)



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Q91. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

The major product [C] of the following reaction sequence will be :

$$CH_2=CH-CHO \xrightarrow[(ii)\ SOCl_2]{(i)\ NaBH_4} [A] \xrightarrow[\text{Anhy. } AlCl_3]{} [B] \xrightarrow{DBr} [C]$$

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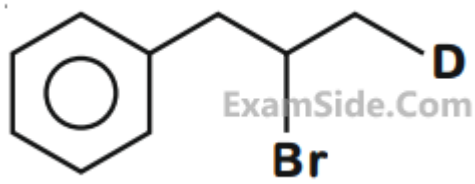
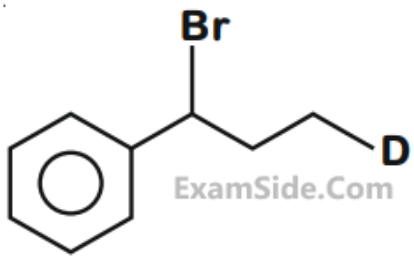
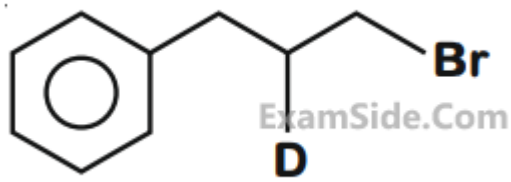
(A)



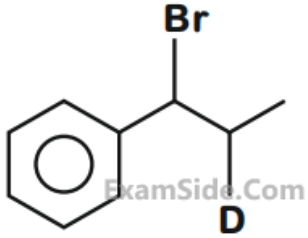
(B)



(C)



(D)



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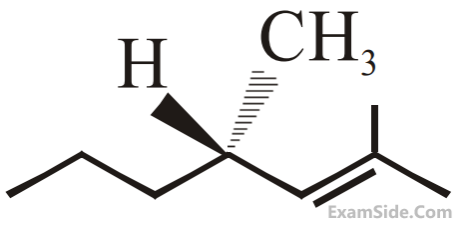
Q92. JEE Main 2020 (Online) 3rd September Morning Slot

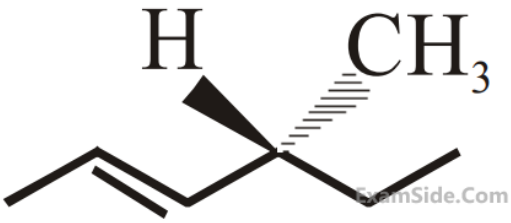
Open solution

Which of the following compounds produces an optically inactive compound on hydrogenation?

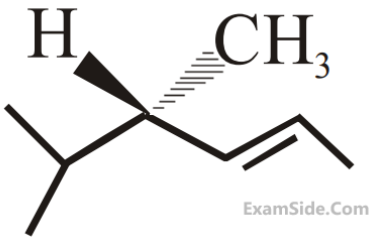
(A)



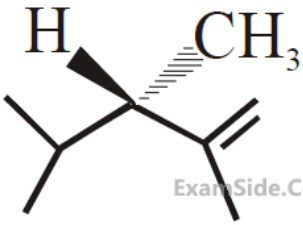
(B)



(C)



(D)



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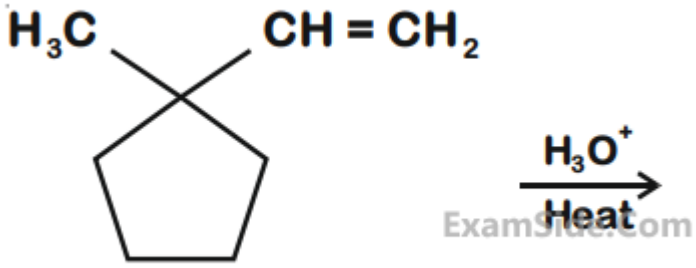
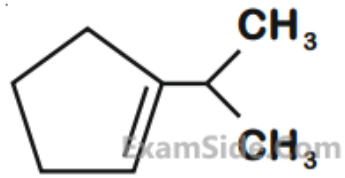
Q93. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

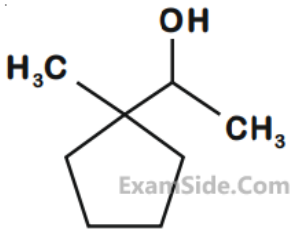
The major product in the following reaction is :



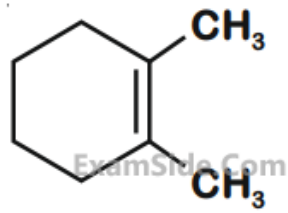
(A)



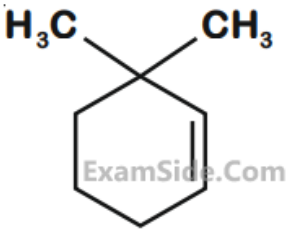
(B)



(C)



(D)



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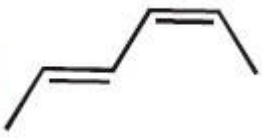
Q94. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

The correct order of heat of combustion for following alkadienes is :

(a)

(b) 

(c)

(A) (b) < (c) < (a) **(B)** (a) < (c) < (b) **(C)** (a) < (b) < (c) **(D)** (c) < (b) < (a)

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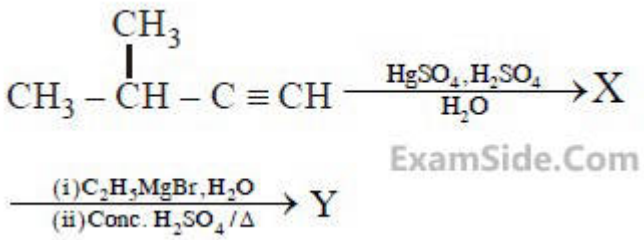
Q95. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

The major product (Y) in the following reactions is :



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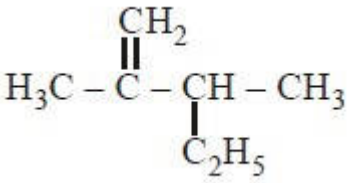
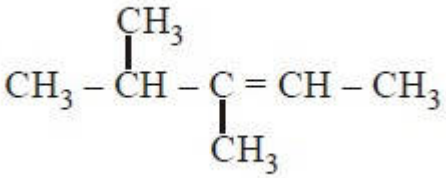
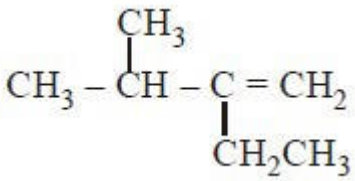
(A)



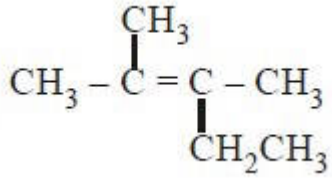
(B)



(C)



(D)



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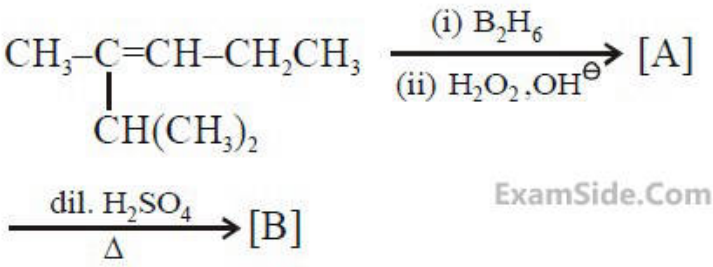
Q96. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

The major product [B] in the following sequence of reactions is :-



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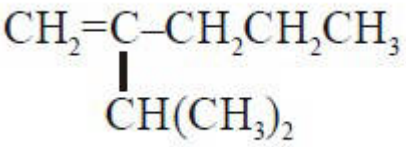
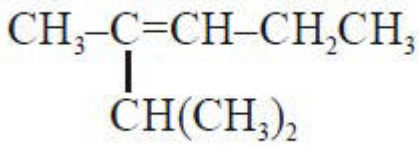
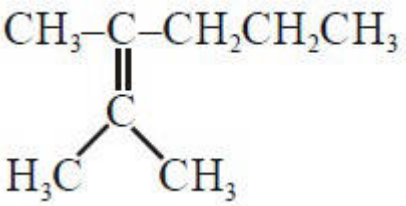
(A)



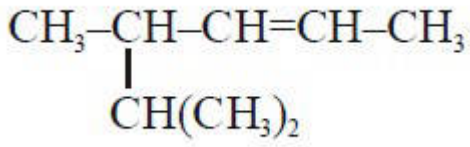
(B)



(C)



(D)



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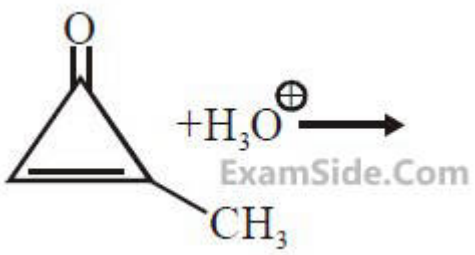
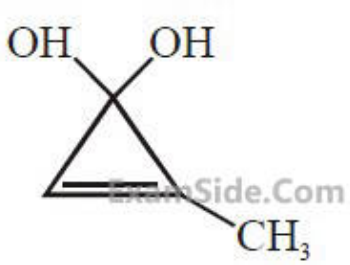
Q97. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

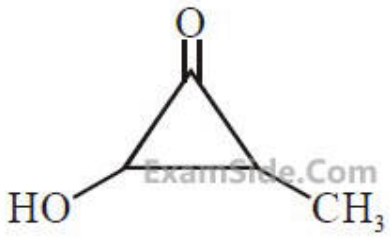
The major product in the following reaction is:



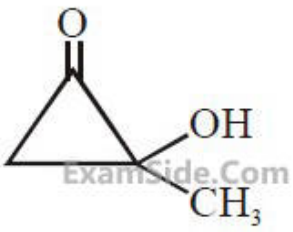
(A)



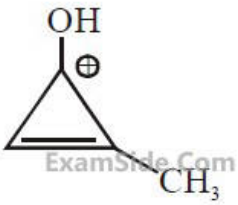
(B)



(C)



(D)



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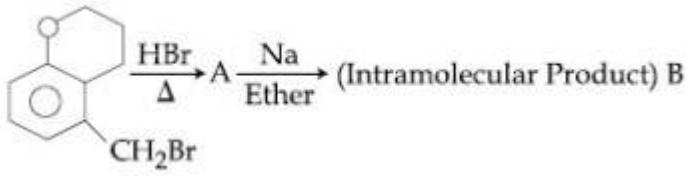
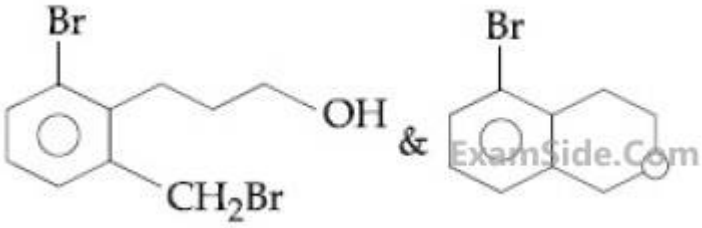
Q98. JEE Main 2020 (Online) 7th January Evening Slot

Open solution

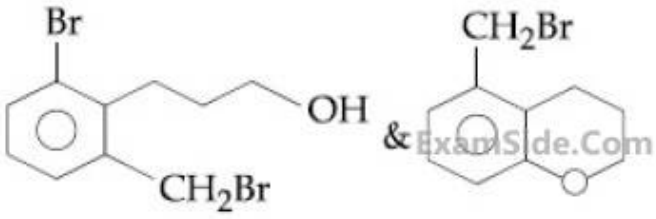
In the following reactions sequence, structure of A and B respectively will be :



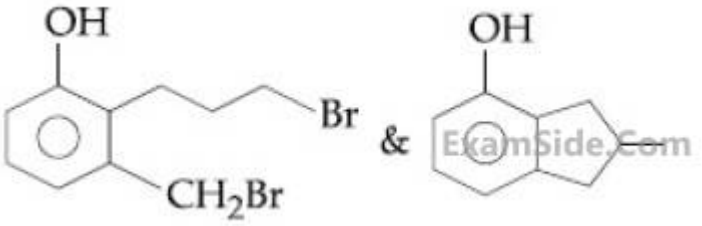
(A)



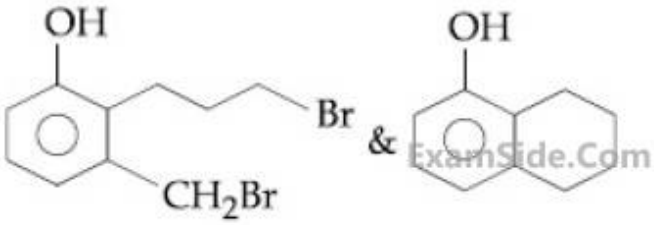
(B)



(C)



(D)



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Q99. JEE Main 2020 (Online) 7th January Morning Slot

Open solution

Consider the following reaction :

$(CH_3)_3CCH(OH)CH_3 \xrightarrow{conc.\ H_2SO_4}$

$(CH_3)_2CHCH(Br)CH_3 \xrightarrow{alc.\ KOH}$

$(CH_3)_2CHCH(Br)CH_3 \xrightarrow{(CH_3)_3O^{\ominus}K^{\oplus}}$

$(CH_3)_2\underset{\underset{OH}{|}}{C}-CH_2-CHO \xrightarrow{\Delta}$

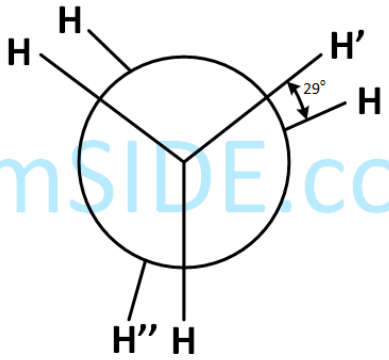
Which of these reaction(s) will not produce Saytzeff product ?

(A) (d) only **(B)** (b) and (d) **(C)** (a), (c) and (d) **(D)** (c) only

Q100. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

In the following skew conformation of ethane, H'–C–C–H" dihedral angle is :



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(A) 120° **(B)** 58° **(C)** 151° **(D)** 149°

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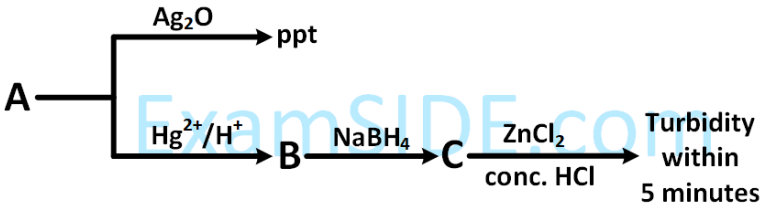
Q101. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

Consider the following reactions :



'A' is :

(A) $CH \equiv CH$

(B) $CH_3 - C \equiv C - CH_3$

(C) $CH_3 - C \equiv CH$

(D) $CH_2 = CH_2$

Q102. JEE Main 2019 (Online) 12th April Morning Slot

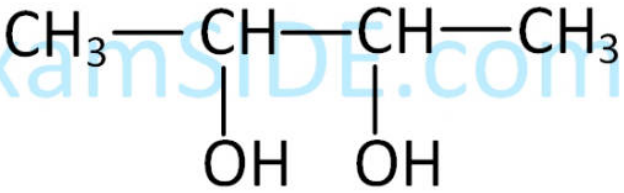
Open solution

But-2-ene on reaction with alkaline $KMnO_4$ at elevated temperature followed by acidification will give :

(A) one molecule of CH_3CHO and one molecule of CH_3COOH

(B) 2 molecules of CH_3COOH

(C)



(D) 2 molecules of CH_3CHO

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Q103. JEE Main 2019 (Online) 12th April Morning Slot

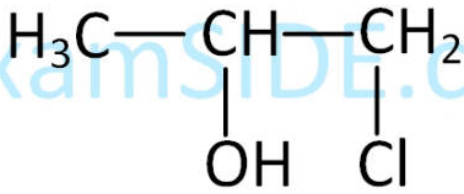
Open solution

The major product of the following addition reaction is :

$H_3C-CH=CH_2 \xrightarrow{Cl_2/H_2O}$

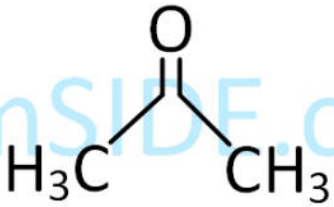
(A)

(B)



(C)

(D)



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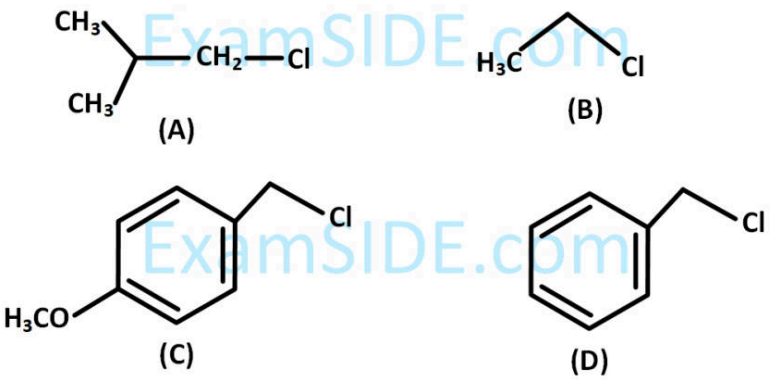
Q104. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

Increasing order of reactivity of the following compounds for S_N1 substitution is:



(A) (A) < (B) < (D) < (C)

(B) (B) < (C) < (D) < (A)

(C) (B) < (A) < (D) < (C)

(D) (B) < (C) < (A) < (D)

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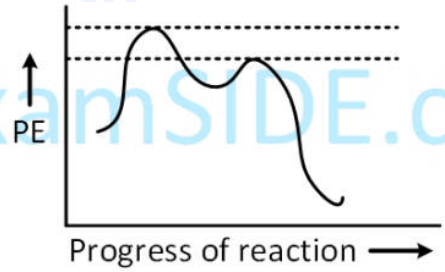
Q105. JEE Main 2019 (Online) 9th April Evening Slot

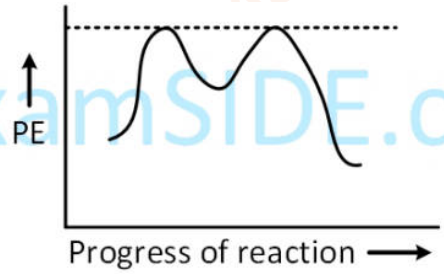
Open solution

Which of the following potential energy (PE) diagrams represents the S_N1 reaction?

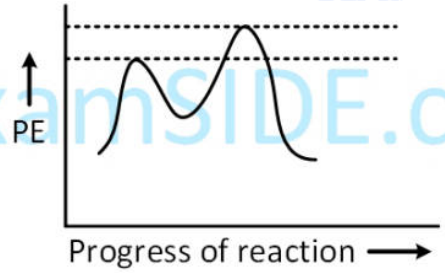
(A)



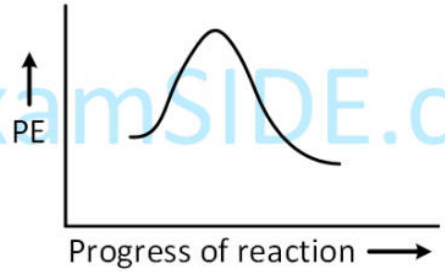
(B)



(C)



(D)



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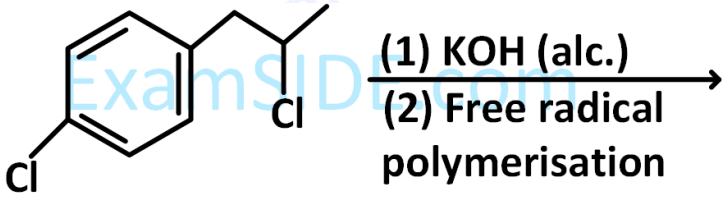
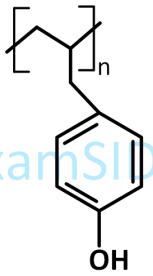
Q106. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

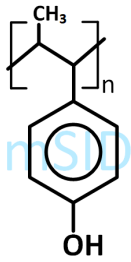
The major product of the following reaction is :



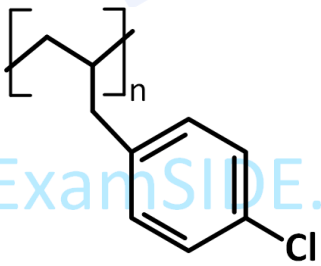
(A)



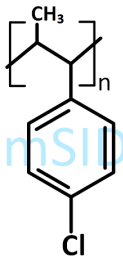
(B)



(C)



(D)



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Q107. JEE Main 2019 (Online) 9th April Morning Slot

The major product of the following reaction is :

Open solution

(A) $CH_3CD_2CH(Cl)(I)$ (B) $CH_3CD(Cl)CHD(I)$ (C) $CH_3C(I)(Cl)CHD_2$ (D) $CH_3CD(I)CHD(Cl)$

Q108. JEE Main 2019 (Online) 8th April Evening Slot

Polysubstitution is a major drawback in:

(A) Reimer Tiemann reaction
(B) Friedel Craft's alkylation
(C) Acetylation of aniline
(D) Friedel Craft's acylation

Open solution

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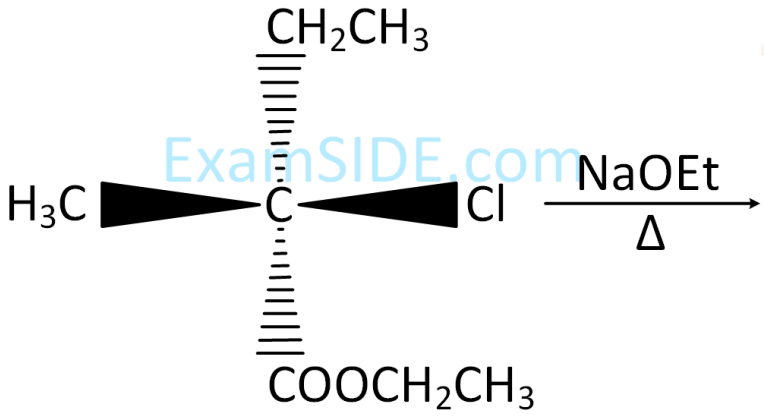
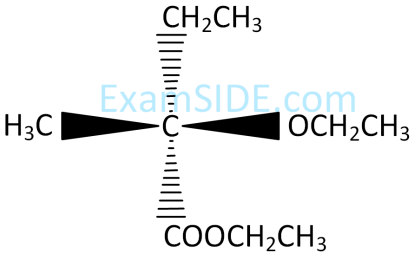
Q109. JEE Main 2019 (Online) 12th January Evening Slot

The major product of the following reaction is –

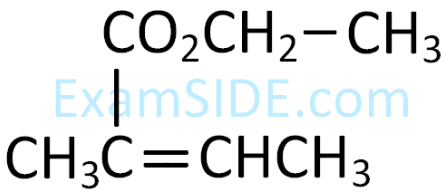
Open solution



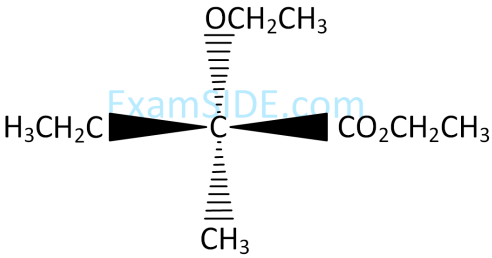
(A)



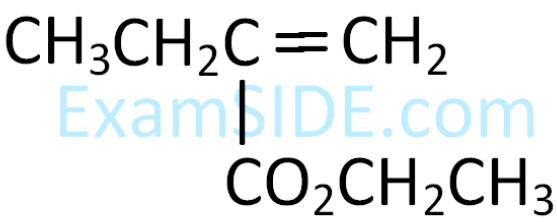
(B)



(C)



(D)



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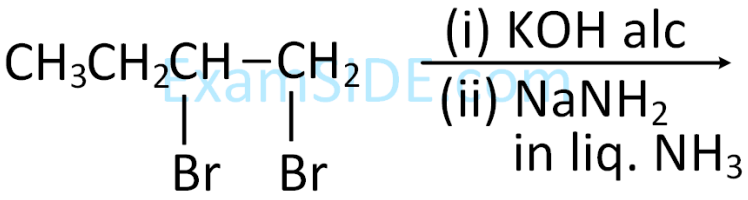
Q110. JEE Main 2019 (Online) 12th January Evening Slot

The major product of the following reaction is :

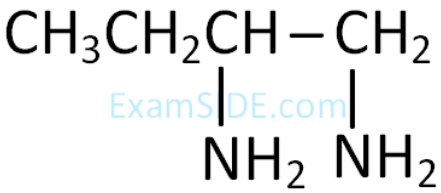
Open solution



(A) $CH_3CH_2C{\equiv}CH$

(B) $CH_3CH{=}C{=}CH_2$

(C)



(D) $CH_3CH{=}CHCH_2NH_2$

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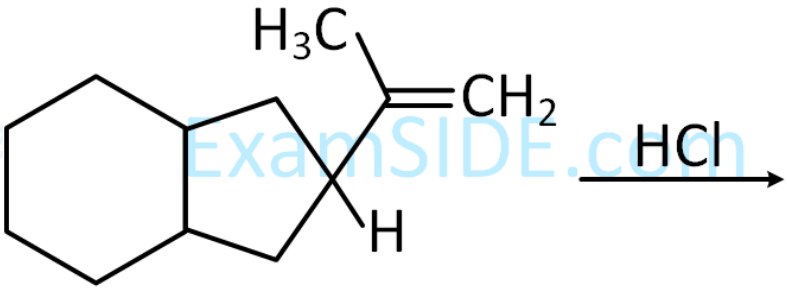
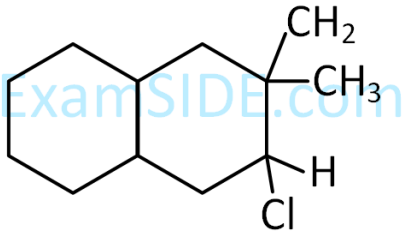
Q111. JEE Main 2019 (Online) 12th January Evening Slot

The major product of the following reaction is –

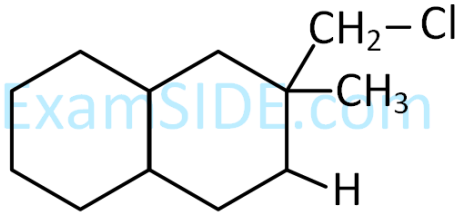
Open solution



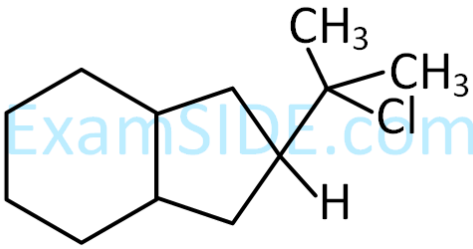
(A)



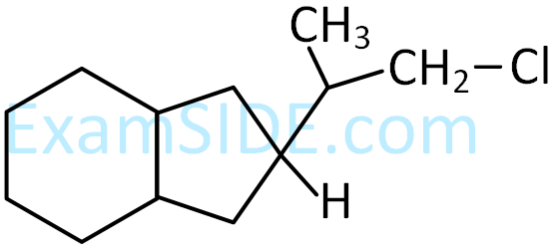
(B)



(C)



(D)



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Q112. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

The correct order for acid strength of compounds :

$CH \equiv CH$, $CH_3-C \equiv CH$ and $CH_2 = CH_2$ is as follows

(A) $CH \equiv CH > CH_2 = CH_2 > CH_3 - C \equiv CH$

(B) $CH_3 - C \equiv CH > CH_2 = CH_2 > HC \equiv CH$

(C) $HC \equiv CH > CH_3 - C \equiv CH > CH_2 = CH_2$

(D) $CH_3 - CH \equiv CH > CH \equiv CH > CH_2 = CH_2$

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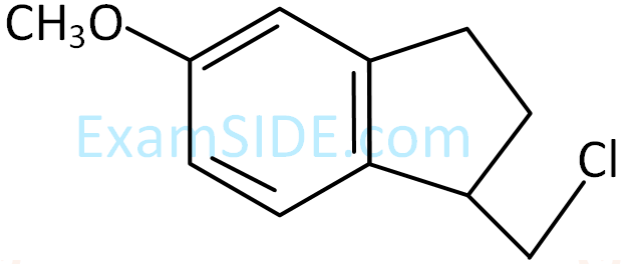
Q113. JEE Main 2019 (Online) 12th January Morning Slot

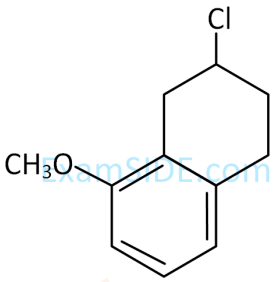
The major product of the following reaction is

Open solution

(A)

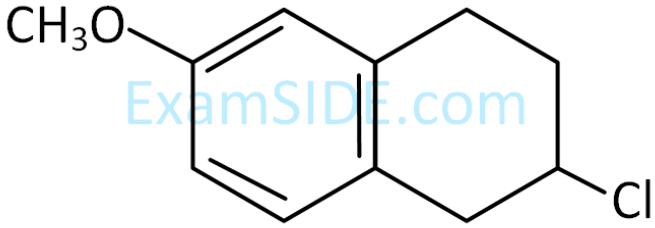


(B)



(C)

(D)



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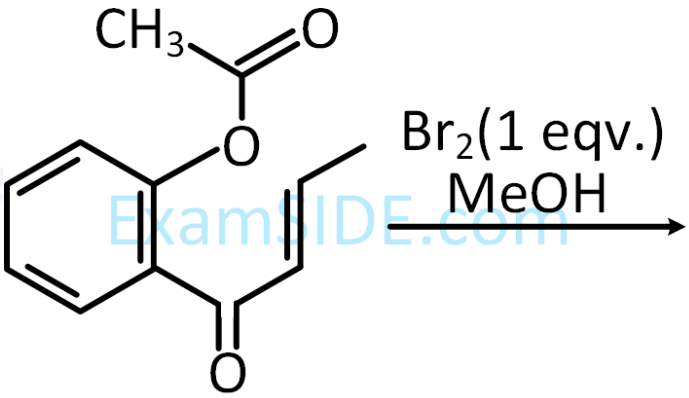
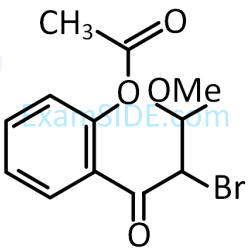
Q114. JEE Main 2019 (Online) 11th January Evening Slot

The major product obtained in the following conversion is

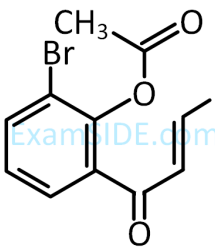


Open solution

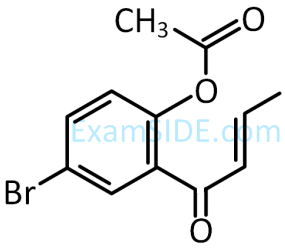
(A)



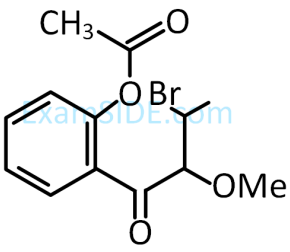
(B)



(C)



(D)



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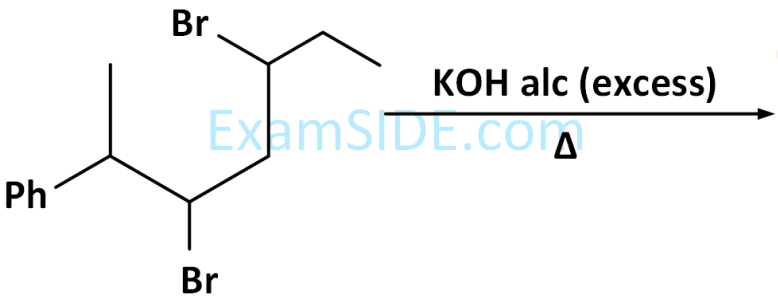
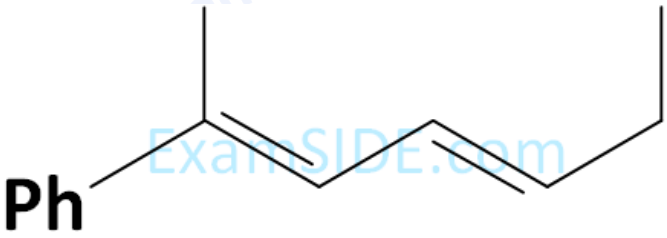
Q115. JEE Main 2019 (Online) 10th January Morning Slot

The major product of the following reaction is

Open solution

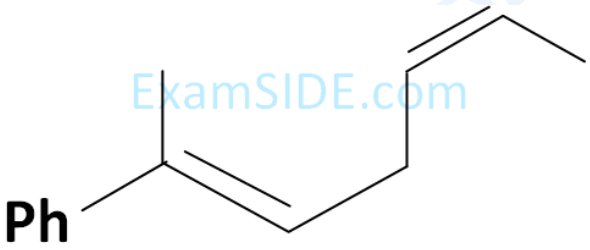


(A)

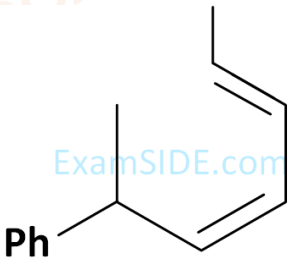


(B)

(C)



(D)



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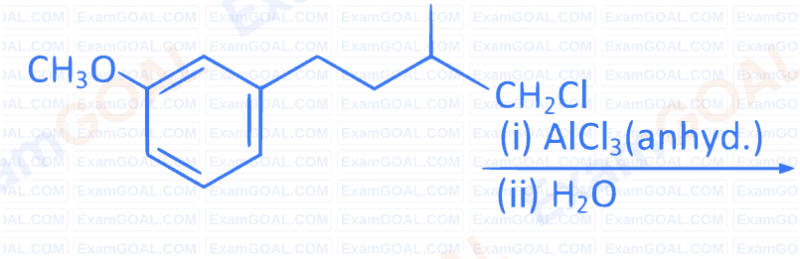
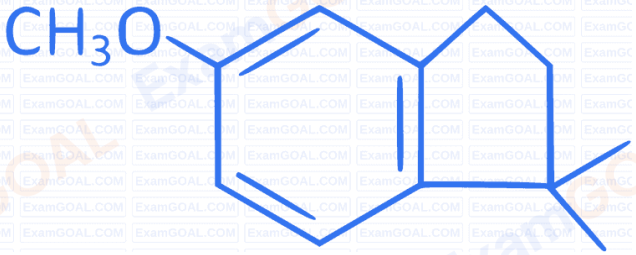
Q116. JEE Main 2019 (Online) 10th January Morning Slot

Open solution

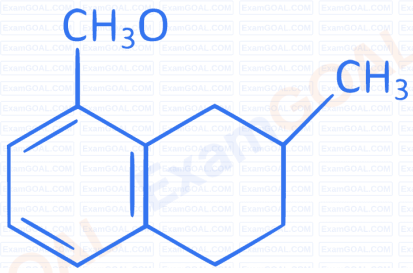
The major product of the following reaction is :



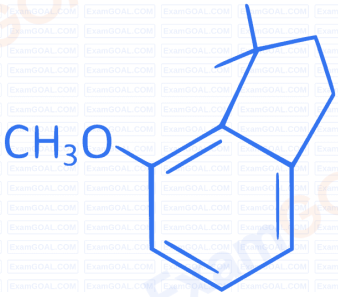
(A)



(B)



(C)



(D)

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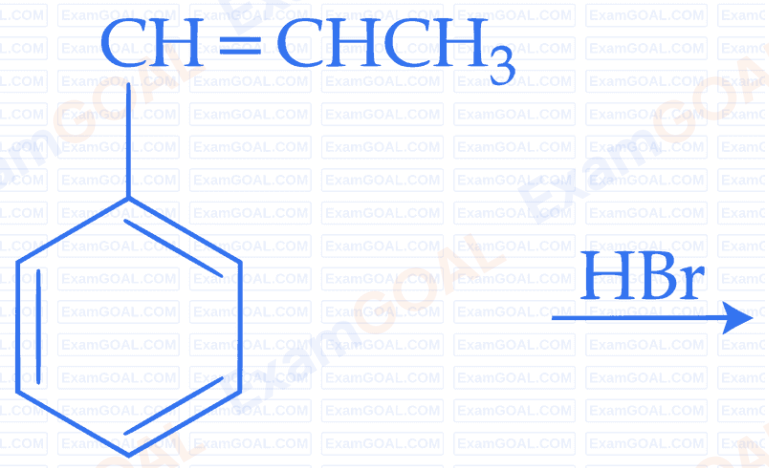
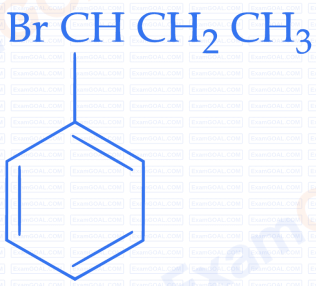
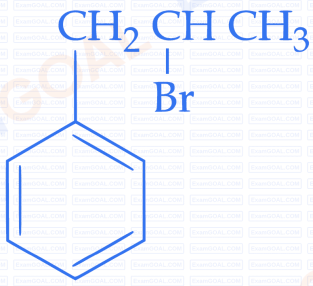
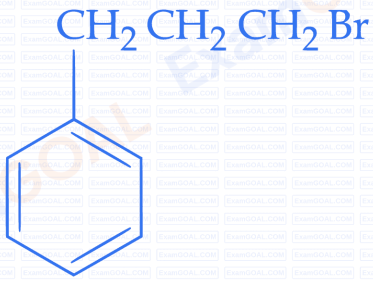
Q117. JEE Main 2018 (Online) 16th April Morning Slot

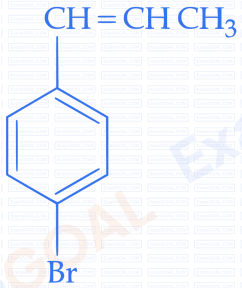
The major product of the following reaction is :



(A) 

(B) 

(C) 

(D) 

Open solution

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Q118. JEE Main 2018 (Offline)

The trans-alkenes are formed by the reduction of alkynes with

(A) Sn - HCl (B) H_2 – Pd/C, $BaSO_4$ (C) $NaBH_4$ (D) Na/liq. NH_3

Open solution

Q119. JEE Main 2018 (Online) 15th April Evening Slot

When 2-butyne is treated with H_2/Lindlar's catalyst, compound X is produced as the major product and when treated with Na/liq. NH_3 it produces Y as the major product. Which of the following statements is correct ?

(A) X will have higher dipole moment and higher boiling point than Y.

(B) Y will have higher dipole moment and higher boiling point than X.

(C) X will have lower dipole moment and lower boiling point than Y.

(D) Y will have higher dipole moment and lower boiling point than X.

Open solution

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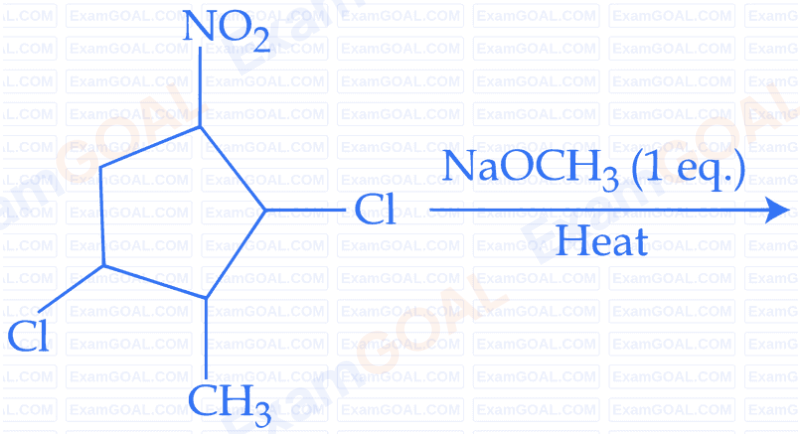
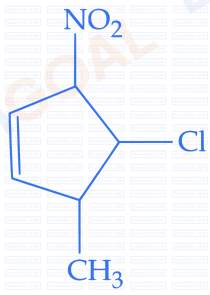
Q120. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

The major product formed in the following reaction is :

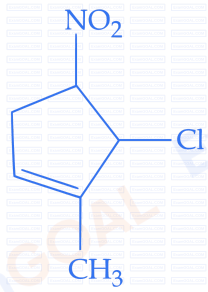


(A)

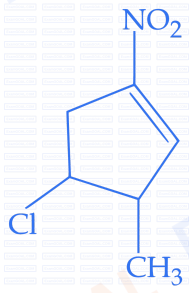


(B)

(C)



(D)



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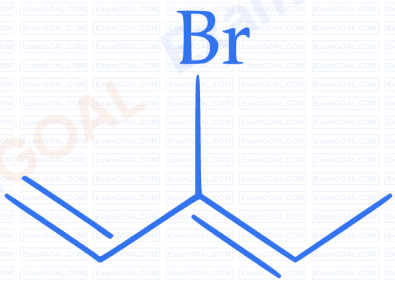
Q121. JEE Main 2018 (Online) 15th April Morning Slot

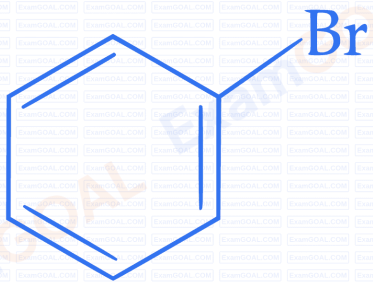
Open solution

Which of the following will most readily give the dehydrohalogenation product ?

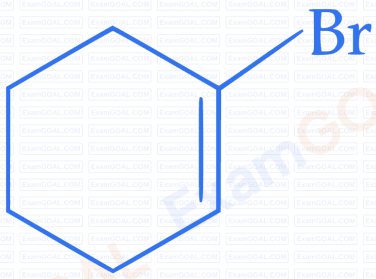
(A)



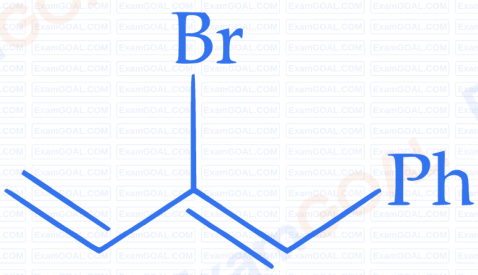
(B)



(C)



(D)



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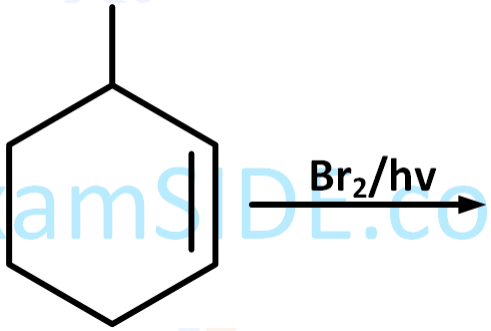
Q122. JEE Main 2017 (Online) 9th April Morning Slot

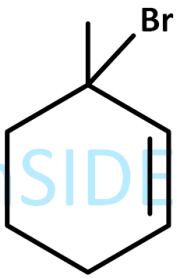
Open solution

The major product of the following reaction is :

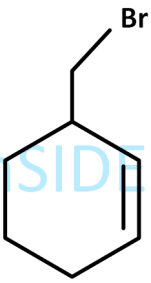


(A)

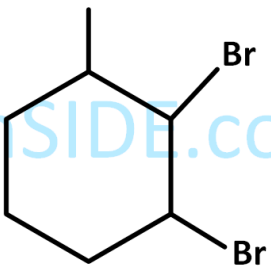
(B)



(C)



(D)



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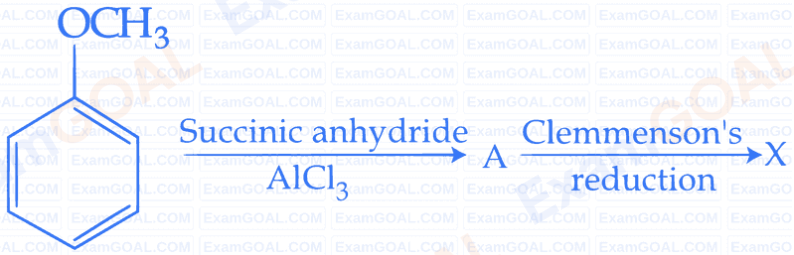
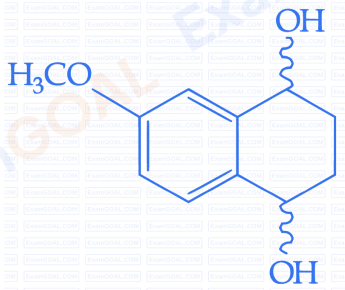
Q123. JEE Main 2016 (Online) 10th April Morning Slot

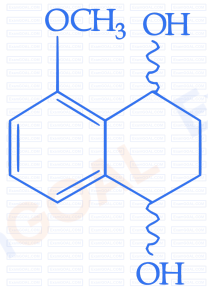
Open solution

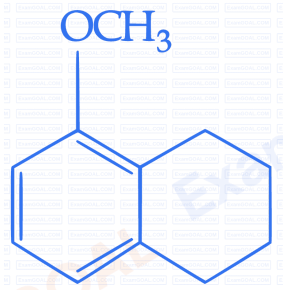
Consider the reaction sequence below :



X is:

(A) 

(B) 

(C) 

(D)

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Q124. JEE Main 2016 (Online) 9th April Morning Slot

5 L of an alkane requires 25 L of oxygen for its complete combustion. If all volumes are measured at constant temperature and pressure, the alkane is :

(A) Ethane (B) Propane (C) Butane (D) Isobutane

Open solution

Q125. JEE Main 2016 (Offline)

The reaction of propene with HOCl ($Cl_2 + H_2O$) proceeds through the intermediate:

(A) $CH_3 - CH^+ - CH_2 - OH$

(B) $CH_3 - CHCl - CH_2^+$

(C) CH_3 - CH^+ - CH_2 - OH

(D) $CH_3 - CH^+ - CH_2 - Cl$

Open solution

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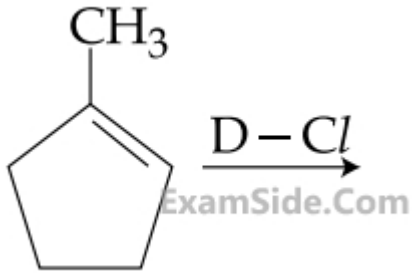
Q126. JEE Main 2015 (Online) 11th April Morning Slot

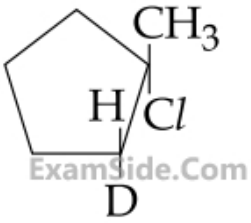
Open solution

What is the major product expected from the following reaction ?

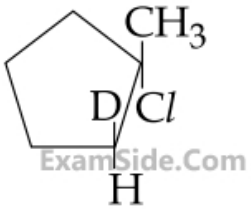


Where D is an isotope of Hydrogen.

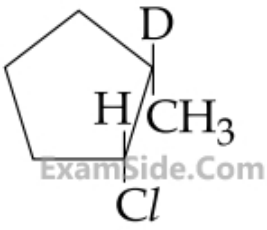
(A)



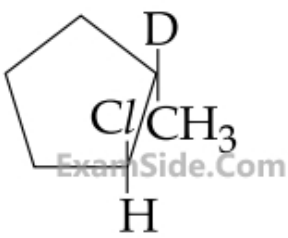
(B)



(C)



(D)



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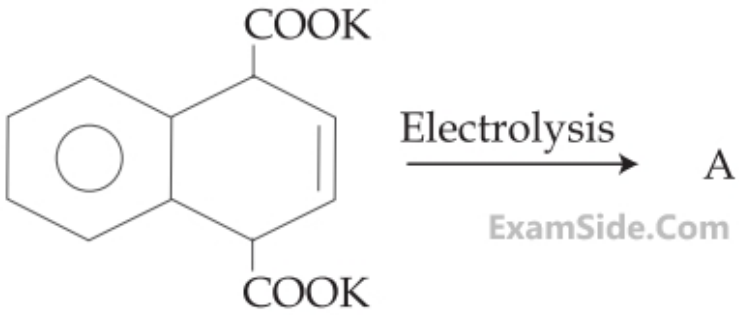
Q127. JEE Main 2015 (Online) 10th April Morning Slot

Open solution



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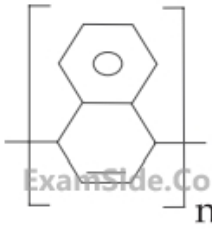
A is :

(A)

(B)

(C)

(D)



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Q128. JEE Main 2014 (Online) 12th April Morning Slot

In the presence of peroxide, HCl and HI do not give anti-Markownikoff's addition to alkenes because :

(A) One of the steps is endothermic in HCl and HI

(B) Both HCl and HI are strong acids

(C) HCl is oxidizing and the HI is reducing

(D) All the steps are exothermic in HCl and HI

Open solution

Q129. JEE Main 2014 (Online) 12th April Morning Slot

The major product obtained in the photo catalysed bromination of 2-methylbutane is :

(A) 1-bromo-2-methylbutane

(B) 1-bromo-3-methylbutane

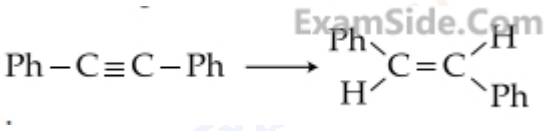
(C) 2-bromo-3-methylbutane

(D) 2-bromo-2-methylbutane

Open solution

Q130. JEE Main 2014 (Online) 11th April Morning Slot

The reagent needed for converting



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is :

(A) Cat. Hydrogenation

(B) H_2/Lindlar Cat.

(C) Li/NH_3

(D) $LiAlH_4$

Open solution

Q131. JEE Main 2014 (Online) 11th April Morning Slot

The gas liberated by the electrolysis of Dipotassium succinate solution is :

(A) Ethane (B) Ethyne (C) Ethene (D) Propene

Open solution

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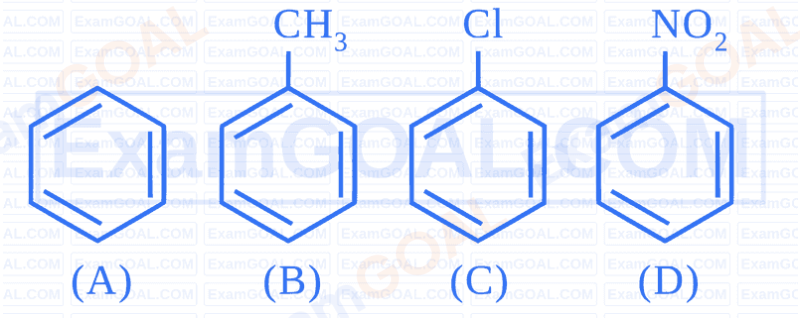
Q132. JEE Main 2013 (Online) 25th April Morning Slot

Open solution

Given



In the above compounds correct order of reactivity in electrophilic substitution reactions will be:

(A) B > A > C > D **(B)** D > C > B > A **(C)** A > B > C > D **(D)** B > C > A > D

Q133. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

Which of the following would not give 2-phenylbutane as the major product in a Friedel-Crafts alkylation reaction?

(A) 1-butene + HF **(B)** 2-butanol + H_2SO_4

(C) Butanoyl chloride + $AlCl_3$ then Zn, HCl **(D)** Butyl chloride + $AlCl_3$

Q134. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

The addition of HI in the presence of peroxide catalyst does not follow anti-Markovnikov's rule because :

(A) HI is a strong reducing agent.

(B) H-I bond is too strong to be broken homolytically.

(C) I atom combines with H atom to give back HI.

(D) Iodine atom is not reactive enough to add across a double bond.

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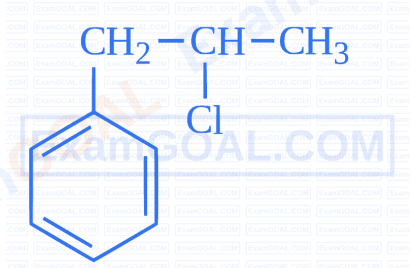
Q135. JEE Main 2013 (Online) 9th April Morning Slot

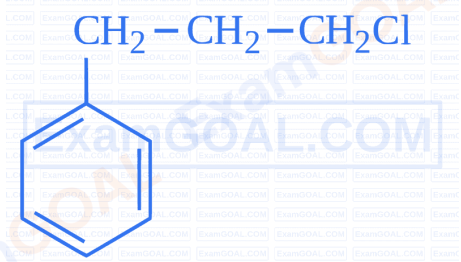
Open solution

X is:

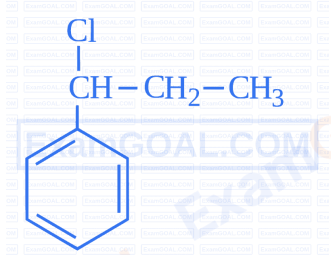
(A)



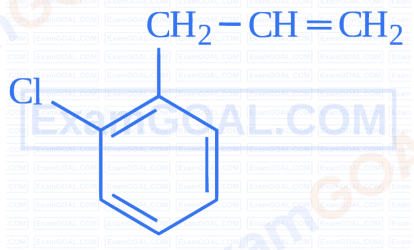
(B)



(C)



(D)



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Q136. AIEEE 2012 (Online) 26th May

The hydration of propyne results in formation of

(A) Acetone (B) Propanol-1 (C) Propene (D) Propanal

Open solution

Q137. AIEEE 2012 (Online) 19th May

The major product obtained in the photobromination of 2-methyl butane is

(A) 2-bromo-2-methylbutane (B) 1-bromo-2-methylbutane

(C) 1-bromo-3-methylbutane (D) 2-bromo-3-methylbutane

Open solution

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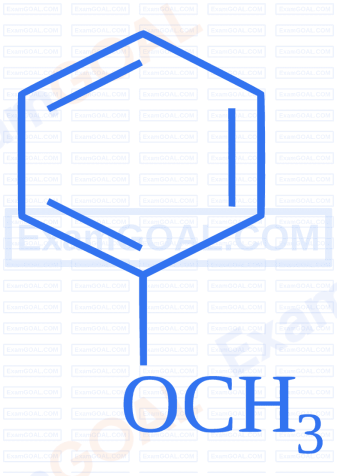
Q138. AIEEE 2012 (Online) 12th May

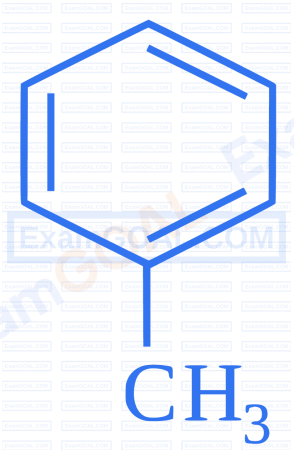
Open solution

In the below mentioned compounds the decreasing order of reactivity towards electrophilic substitution is

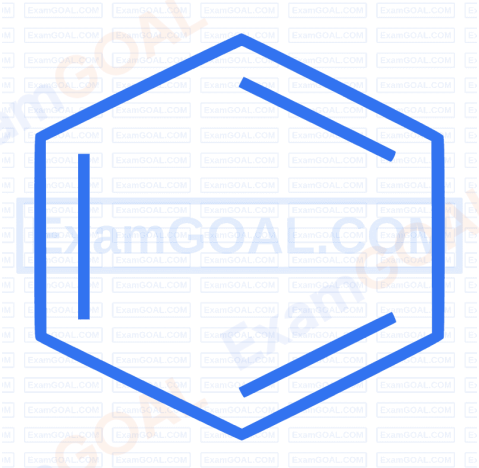
(i)



(ii)



(iii)



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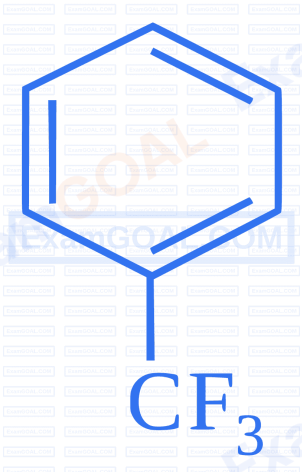
(iv)



(A) $(\text{iv}) > (\text{i}) > (\text{ii}) > (\text{iii})$

(B) $(\text{ii}) > (\text{iii}) > (\text{i}) > (\text{iv})$

(C) $(\text{iii}) > (\text{i}) > (\text{iv}) > (\text{ii})$

(D) $(\text{i}) > (\text{ii}) > (\text{iii}) > (\text{iv})$

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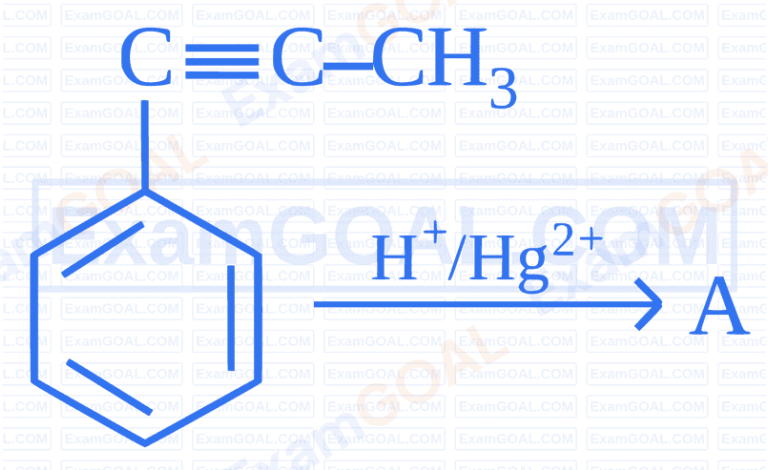
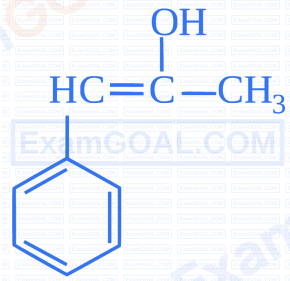
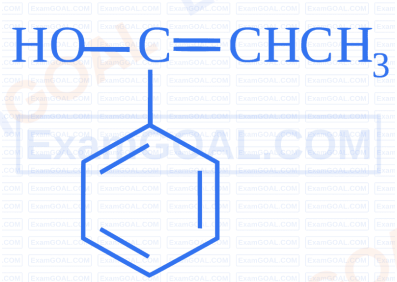
Q139. AIEEE 2012 (Online) 12th May

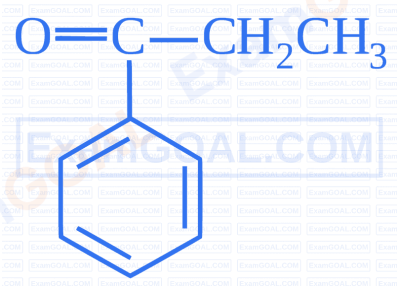


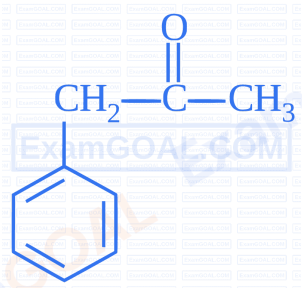
Open solution

In the given reaction, the product 'A' is

(A) 

(B) 

(C) 

(D) 

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Q140. AIEEE 2012

Which branched chain isomer of the hydrocarbon with molecular mass 72u gives only one isomer of mono substituted alkyl halide ?

(A) Tertiary butyl chloride
(B) Neopentane
(C) Isohexane
(D) Neohexane

Open solution

Q141. AIEEE 2012

2–Hexyne gives trans–2–Hexene on treatment with :

(A) Pt/H_2
(B) Li/NH_3
(C) $Pd/BaSO_4$
(D) $LiAlH_4$

Open solution

Q142. AIEEE 2011

Ozonolysis of an organic compound gives formaldehyde as one of the products. This confirms the presence of :

(A) a vinyl group
(B) an isopropyl group
(C) an acetylenic triple bond
(D) two ethylenic double bonds

Open solution

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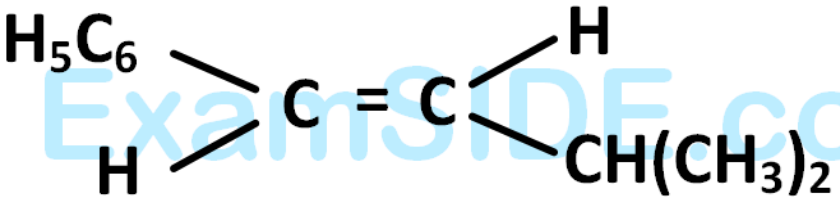
Q143. AIEEE 2010

The main product of the following reaction is

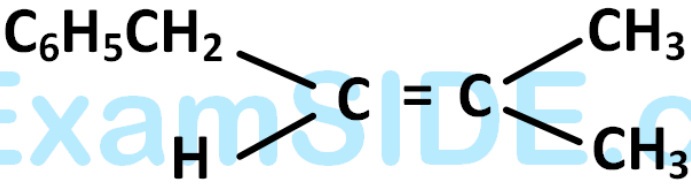
$$C_6H_5CH_2CH(OH)CH(CH_3)_2 \xrightarrow{conc.H_2SO_4} ?$$

Open solution

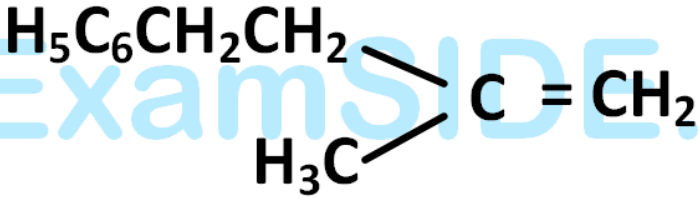
(A)



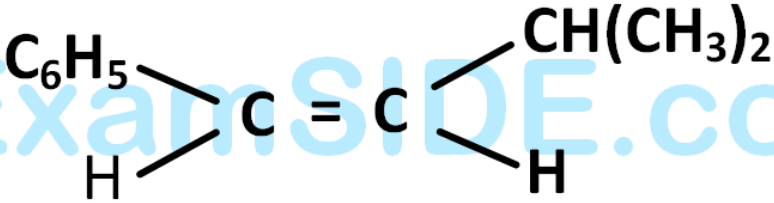
(B)



(C)



(D)



Q144. AIEEE 2008

Toluene is nitrated and the resulting product is reduced with tin and hydrochloric acid. The product so obtained is diazotised and then heated with cuprous bromide. The reaction mixture so formed contains

Open solution

(A) mixture of o– and p–bromotoluenes

(B) mixture of o– and p–dibromobenzenes

(C) mixture of o– and p–bromoanilines

(D) mixture of o– and m–bromotoluenes

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Q145. AIEEE 2008

In the following sequence of reactions, the alkene affords the compound 'B'

$CH_3 - CH = CH - CH_3 \xrightarrow{O_3} A \xrightarrow[Zn]{H_2O} B$

The compound B is

(A) CH_3CH_2CHO (B) CH_3COCH_3 (C) $CH_3CH_2COCH_3$ (D) CH_3CHO

Open solution

Q146. AIEEE 2008

The hydrocarbon which can react with sodium in liquid ammonia is

(A) $CH_3CH_2CH_2C{\equiv}CCH_2CH_2CH_3$ (B) $CH_3CH_2C{\equiv}CH$

(C) $CH_3CH{=}CHCH_3$ (D) $CH_3CH_2C{\equiv}CCH_2CH_3$

Open solution

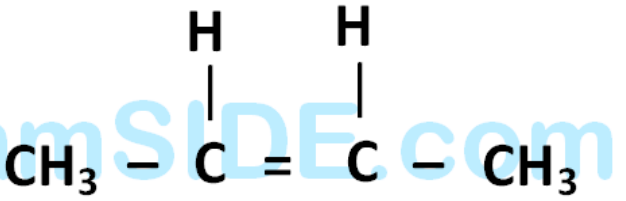
Q147. AIEEE 2008

The treatment of CH_3MgX with $CH_3C \equiv C - H$ produces

(A) $CH_3 - CH = CH_2$

(B) $CH_3C \equiv C - CH_3$

(C)



(D) CH_4

Open solution

Q148. AIEEE 2007

Which of the following reactions will yield 2, 2-dibromopropane?

(A) $CH_3 - CH = CH_2 + HBr \rightarrow$

(B) $CH_3 - C \equiv CH + 2HBr \rightarrow$

(C) $CH_3CH = CHBr + HBr \rightarrow$

(D) $CH \equiv CH + 2HBr \rightarrow$

Open solution

Q149. AIEEE 2007

The reaction of toluene with Cl_2 in presence of $FeCl_3$ gives predominantly

(A) benzoyl chloride (B) benzyl chloride

(C) o-and p-chlorotoluene (D) m-chlorotoluene

Open solution

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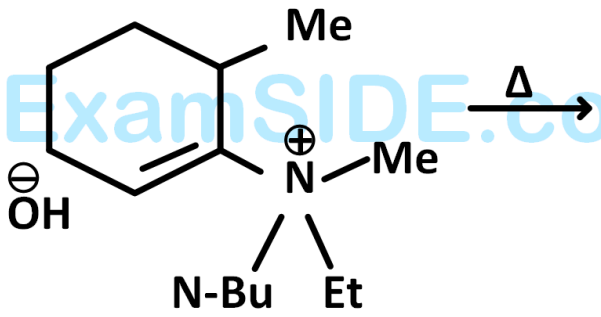
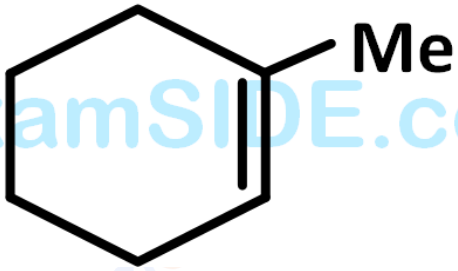
Q150. AIEEE 2006

Open solution

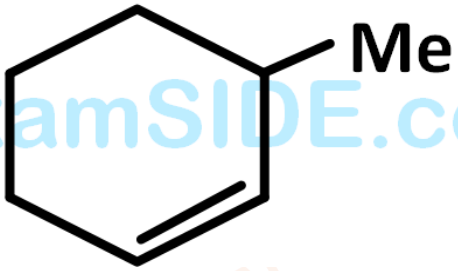


The alkene formed as a major product in the above elimination reaction is

(A)



(B)



(C)

(D)

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Q151. AIEEE 2005

2-methylbutane on reacting with bromine in the presence of sunlight gives mainly

(A) 1-bromo-2-methylbutane
(B) 2-bromo-2-methylbutane
(C) 2-bromo-3-methylbutane
(D) 1-bromo-3-methylbutane

Open solution

Q152. AIEEE 2005

Reaction of one molecule of HBr with one molecule of 1,3-butadiene at 40^{o}C gives predominantly

(A) 3-bromobutene under kinetically controlled conditions
(B) 1-bromo-2-butene under thermodymically controlled conditions
(C) 3-bromobutene under thermodynamically controlled conditions
(D) 1-bromo-2-butene under kinetically controlled conditions

Open solution

Q153. AIEEE 2005

Of the five isomeric hexanes, the isomer which can give two monochlorinated compounds is

(A) n-hexane
(B) 2, 3-dimethylbutane
(C) 2,2-dimethylbutane
(D) 2-methylpentane

Open solution

Q154. AIEEE 2005

Which types of isomerism is shown by 2,3-dichlorobutane?

(A) Diastereo
(B) Optical
(C) Geometric
(D) Structural

Open solution

Q155. AIEEE 2004

Which one of the following has the minimum boiling point?

(A) n-butane
(B) isobutane
(C) 1- butene
(D) 1- butyne

Open solution

Q156. AIEEE 2003

On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be

(A) pentane
(B) isopentane
(C) neopentane
(D) propane

Open solution

Q157. AIEEE 2003

Butene-1 may be converted to butane by reaction with

(A) Sn - HCl
(B) Zn - Hg
(C) Pd/H_2
(D) Zn - HCl

Open solution

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Q158. AIEEE 2002

Which of these will not react with acetylene ?

(A) NaOH (B) ammoniacal $AgNO_3$

(C) Na (D) HCl

Open solution

Q159. AIEEE 2002

What is the product when acetylene reacts with hypochlorous acid ?

(A) CH_3COCl (B) $ClCH_2CHO$ (C) Cl_2CHCHO (D) ClCHCOOH

Open solution

Q160. AIEEE 2002

Acetylene does not react with :

(A) Na (B) ammoniacal $AgNO_3$

(C) HCl (D) $NaOH$

Open solution

Numerical 28 questions

Q161. JEE Main 2026 (Online) 6th April Evening Shift

An alkane (Y) requires 8 moles of oxygen for complete combustion and on chlorination with $Cl_2/h\nu$, (Y) gives only one monochlorinated product (Z). The total number of primary carbon atoms in (Y) is ____ .

Open solution

Q162. JEE Main 2026 (Online) 2nd April Morning Shift

Consider the isomers of hydrocarbon with molecular formula C_5H_{10}. These isomers do not decolourise $KMnO_4$ solution. These isomers are subjected to chlorination with chlorine in presence of light to give monochloro compounds. The total number of monochloro compounds (structural isomers only) formed is _____.

Open solution

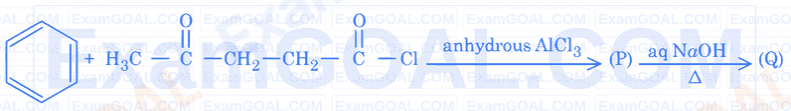
Q163. JEE Main 2026 (Online) 2nd April Morning Shift

One mole of an alkane (x) requires 8 mole oxygen for complete combustion. Sum of number of carbon and hydrogen atoms in the alkane (x) is _____.

Open solution

Q164. JEE Main 2026 (Online) 23rd January Evening Shift

Consider the following reaction of benzene.



In compound (Q), the percentage of oxygen is ____ %. (Nearest integer)

Open solution

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Q165. JEE Main 2026 (Online) 22nd January Morning Shift

The cycloalkene (X) on bromination consumes one mole of bromine per mole of (X) and gives the product (Y) in which $C : Br$ ratio is 3:1. The percentage of bromine in the product (Y) is ____ %. (Nearest integer)

(Given : molar mass in $gmol^{-1} H : 1, C : 12, O : 16, Br : 80$)

Open solution

Q166. JEE Main 2025 (Online) 29th January Evening Shift

Isomeric hydrocarbons → negative Baeyer's test

(Molecular formula C_9H_{12})

The total number of isomers from above with four different non-aliphatic substitution sites is -

Open solution

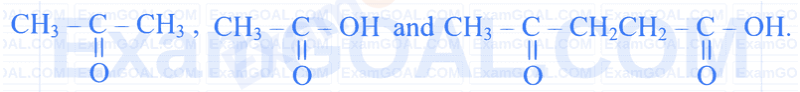
Q167. JEE Main 2025 (Online) 29th January Morning Shift

The sum of sigma (σ) and pi (π) bonds in Hex-1,3-dien-5-yne is _____.

Open solution

Q168. JEE Main 2025 (Online) 23rd January Evening Shift

A compound ' X ' absorbs 2 moles of hydrogen and ' X ' upon oxidation with $KMnO_4 \mid H^+$ gives



The total number of σ bonds present in the compound ' X ' is _____.

Open solution

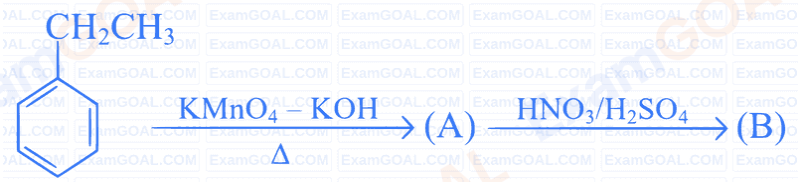
Q169. JEE Main 2025 (Online) 22nd January Evening Shift

The compound with molecular formula C_6H_6, which gives only one monobromo derivative and takes up four moles of hydrogen per mole for complete hydrogenation has _____ π electrons.

Open solution

Q170. JEE Main 2024 (Online) 8th April Morning Shift

Major product B of the following reaction has _____ π-bond.



Open solution

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Q171. JEE Main 2024 (Online) 6th April Morning Shift

The major product of the following reaction is P.

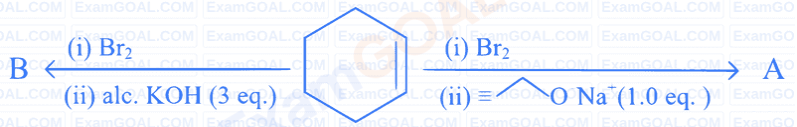
$$CH_3C \equiv C - CH_3 \xrightarrow[\text{(ii) dil. } KMnO_4,\ 273\ K]{\text{(i) Na /liq. } NH_3} \text{'P'}$$

Number of oxygen atoms present in product 'P' is _____. (nearest integer)

Open solution

Q172. JEE Main 2024 (Online) 6th April Morning Shift

The major products from the following reaction sequence are product A and product B.



The total sum of π electrons in product A and product B are _____ (nearest integer)

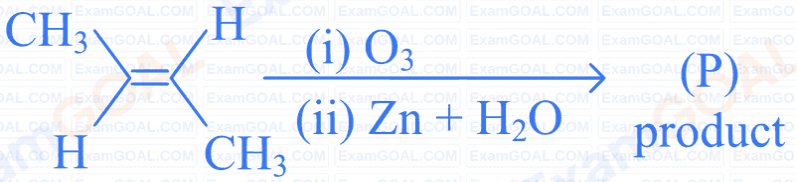
Open solution

Q173. JEE Main 2024 (Online) 31st January Morning Shift

Number of alkanes obtained on electrolysis of a mixture of CH_3COONa and C_2H_5COONa is _____.

Open solution

Q174. JEE Main 2024 (Online) 29th January Morning Shift

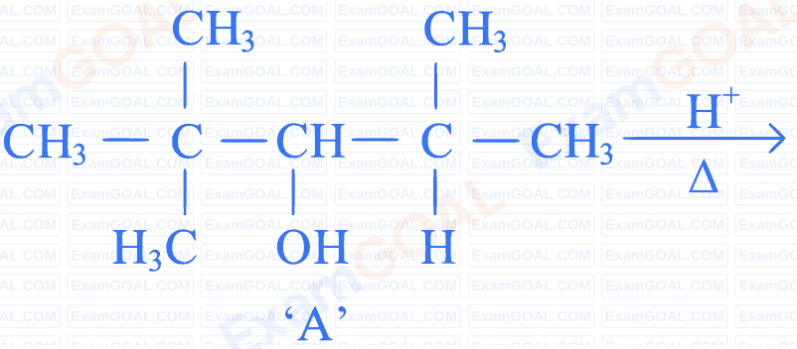


Consider the given reaction. The total number of oxygen atom/s present per molecule of the product (P) is _____.

Open solution

Q175. JEE Main 2023 (Online) 13th April Morning Shift

For the given reaction



The total number of possible products formed by tertiary carbocation of A is _____.

Open solution

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Q176. JEE Main 2023 (Online) 11th April Evening Shift

The number of possible isomeric products formed when 3-chloro-1-butene reacts with HCl through carbocation formation is _____.

Open solution

Q177. JEE Main 2023 (Online) 8th April Morning Shift

Molar mass of the hydrocarbon (X) which on ozonolysis consumes one mole of O_3 per mole of (X) and gives one mole each of ethanal and propanone is _____ $g\ mol^{-1}$

(Molar mass of $C : 12\ g\ mol^{-1}, H : 1\ g\ mol^{-1}$)

Open solution

Q178. JEE Main 2023 (Online) 6th April Morning Shift

Number of bromo derivatives obtained on treating ethane with excess of Br_2 in diffused sunlight is _____

Open solution

Q179. JEE Main 2023 (Online) 29th January Morning Shift

17 mg of a hydrocarbon (M.F. $C_{10}H_{16}$) takes up 8.40 mL of the H_2 gas measured at 0°C and 760 mm of Hg. Ozonolysis of the same hydrocarbon yields

The number of double bond/s present in the hydrocarbon is _____.

Open solution

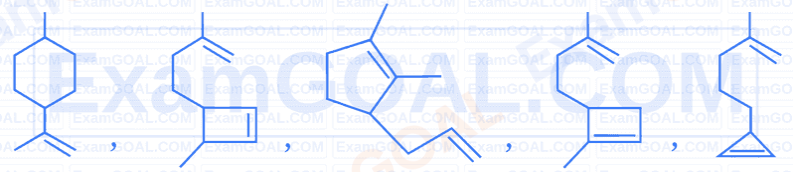
Q180. JEE Main 2023 (Online) 24th January Evening Shift

Maximum number of isomeric monochloro derivatives which can be obtained from 2,2,5,5-tetramethylhexane by chlorination is _____

Open solution

Q181. JEE Main 2022 (Online) 30th July Evening Shift

Among the following, how many hydrocarbon (s) with formula $C_{10}H_{16}$ which on reaction with acidic potassium permagenate yield(s) same product ? _____



Open solution

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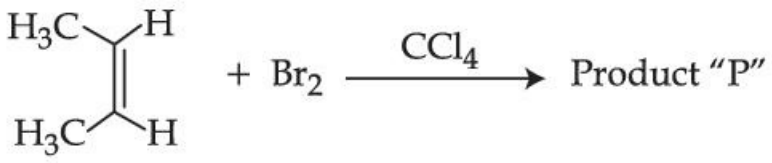
Q182. JEE Main 2022 (Online) 24th June Morning Shift

The major product 'A' of the following given reaction has _____ sp^2 hybridized carbon atoms.

Open solution

Q183. JEE Main 2021 (Online) 25th July Evening Shift

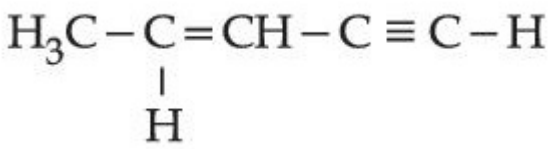


Open solution

Consider the above chemical reaction. The total number of stereoisomers possible for Product 'P' is _____.

Q184. JEE Main 2021 (Online) 25th July Morning Shift

The number of sigma bonds in

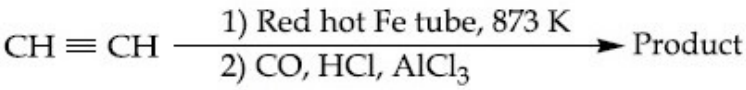


Open solution

is _____.

Q185. JEE Main 2021 (Online) 25th February Morning Shift

Consider the following chemical reaction.



Open solution

The number of sp^2 hybridized carbon atom(s) present in the product is _____.

Q186. JEE Main 2020 (Online) 5th September Morning Slot

The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of butane is ____.

Open solution

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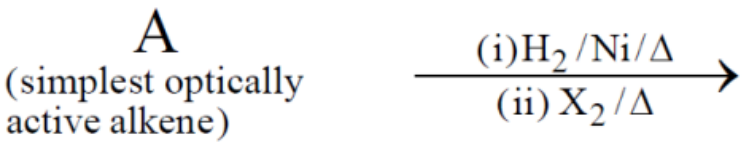
Q187. JEE Main 2020 (Online) 3rd September Morning Slot

The total number of monohalogenated organic products in the following (including stereoisomers) reaction is _____.

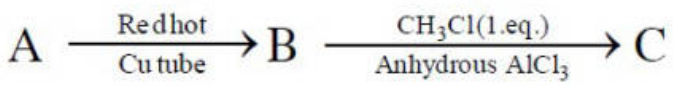
Open solution



Q188. JEE Main 2020 (Online) 8th January Evening Slot

In the following sequence of reactions the maximum number of atoms present in molecule 'C' in one plane is _____.

Open solution



(A is a lowest molecular weight alkyne)

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **B**	3. **D**	4. **A**	5. **C**
6. **B**	7. **B**	8. **D**	9. **C**	10. **C**
11. **D**	12. **A**	13. **A**	14. **A**	15. **D**
16. **A**	17. **C**	18. **D**	19. **C**	20. **B**
21. **A**	22. **B**	23. **D**	24. **C**	25. **D**
26. **B**	27. **C**	28. **A**	29. **D**	30. **D**
31. **D**	32. **D**	33. **A**	34. **B**	35. **D**
36. **B**	37. **D**	38. **D**	39. **C**	40. **D**
41. **D**	42. **A**	43. **B**	44. **A**	45. **A**
46. **D**	47. **A**	48. **B**	49. **A**	50. **B**
51. **D**	52. **D**	53. **B**	54. **A**	55. **A**
56. **D**	57. **A**	58. **B**	59. **B**	60. **B**
61. **C**	62. **D**	63. **D**	64. **D**	65. **C**
66. **D**	67. **A**	68. **B**	69. **A**	70. **C**
71. **B**	72. **A**	73. **D**	74. **A**	75. **B**
76. **D**	77. **D**	78. **D**	79. **B**	80. **C**
81. **C**	82. **A**	83. **A**	84. **B**	85. **A**
86. **D**	87. **D**	88. **B**	89. **A**	90. **C**
91. **B**	92. **D**	93. **C**	94. **C**	95. **D**
96. **C**	97. **D**	98. **D**	99. **D**	100. **D**
101. **C**	102. **B**	103. **B**	104. **C**	105. **A**
106. **D**	107. **C**	108. **B**	109. **B**	110. **A**
111. **C**	112. **C**	113. **D**	114. **A**	115. **A**
116. **A**	117. **A**	118. **D**	119. **A**	120. **D**
121. **D**	122. **B**	123. **D**	124. **B**	125. **D**
126. **B**	127. **B**	128. **A**	129. **D**	130. **C**
131. **C**	132. **A**	133. **C**	134. **D**	135. **C**
136. **A**	137. **A**	138. **D**	139. **C**	140. **B**
141. **B**	142. **A**	143. **A**	144. **A**	145. **D**
146. **B**	147. **D**	148. **B**	149. **C**	150. **B**

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151. **B**	152. **B**	153. **B**	154. **B**	155. **B**
156. **C**	157. **C**	158. **A**	159. **C**	160. **D**

Numerical

161. **4**	162. **14**	163. **17**	164. **10**	165. **66**
166. **2**	167. **15**	168. **27**	169. **8 or 6**	170. **5**
171. **2**	172. **8**	173. **3**	174. **1**	175. **4**
176. **4**	177. **70**	178. **9**	179. **3**	180. **3**
181. **2**	182. **2**	183. **2**	184. **10**	185. **7**
186. **18**	187. **8**	188. **13**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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ANALYSIS

Know your score. Know your rank.

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- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

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